TUNISIAN REUBLIC
Ministry of Higher Education and Scientific Research
University of Carthage
National Institute of Applied Sciences and Technology

Graduation Project

to obtain the

National Engineering Diploma
in Applied Sciences and Technology

Specialty: Instrumentation and Industrial Maintenance

Title
Oil Site Telemetry System Upgrade

Prepared by

Fekri BEN FATMA

Hosted by INTEGRATION OBJECTS

Defended on the 5th of July 2021 in front of a jury composed of:

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Mr. Med Chaker ZAGHDOUDI	Reviewer
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Ms. Imen HARBAOUI	Supervisor at INSAT

School year: 2020 / 2021

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Dedication

The completion of this project wouldn't be possible without the help and encouragement of a lot of people to whom I dedicate this work.

To my **mother**
You have been my motivation and inspiration from the day you gave birth to me. I can not thank you enough for being by my side.

To my **father**
Without you I would not be what I have become today. Your presence gives me courage and dedication to move forward.

To my **sister**
Thank you for being my shield and guardian angel.

To my **roommates**
Well we've had some crazy years together. Been through thick and thin. I couldn't have asked for better roommates.

To all my family and friends
Thanks to everyone I have known for being part of my life. Thank you for your support, trust, and love.

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- Dear members of the jury for accepting to evaluate my work.

- The whole team of "Integration objects" for their friendly welcome and professional cooperation.

- My thanks go finally to everyone who contributed directly or indirectly to the development of this work.

Abstract

The main goal of this project is the upgrade of the telemetry system at an oil site. In this report we describe the necessary steps for implementing the system. Starting with the hardware selection and sizing, followed by the generation of the wiring diagrams and schematics. To finally end with the control logic configuration and the Human-Machine interfaces development.

Keywords: Telemetry system, RTU, MTU, SCADA, Modbus, DNP3, ControlWave Micro, ControlWave RED

Résumé

Le but principal de ce projet est la mise à niveau du système de télémétrie d'un site pétrolier. Dans ce rapport nous exposons les étapes nécessaires pour l'implémentation du système. En commençant par la sélection et le dimensionnement du matériel, suivi par la génération des diagrammes de câblage électrique. Pour finir avec l'implémentation de la logique de commande et le développement des interfaces Homme-Machine.

Keywords: Système de télémétrie, RTU, MTU, SCADA, Modbus, DNP3, ControlWave Micro, ControlWave RED

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List of acronyms

- **OPC.** Open Platform Communication
- **ICS.** Industrial Control Systems
- **SCADA.** Supervisory Control And Data Acquisition
- **DCS.** Distributed Control Systems
- **HMI.** Human Machine Interface
- **PLC.** Programmable Logic Controller
- **RTU.** Remote Telemetry Unit
- **MTU.** Master Telemetry Unit
- **I/O.** Input Output
- **CPU.** Central Processing Unit
- **IEC.** International Electrotechnical Commission
- **IL.** Instruction List
- **ST.** Structured Text
- **LD.** Ladder Diagram
- **FBD.** Function Block Diagram
- **SFC.** Sequential Function Chart
- **UA.** Unified Architecture
- **DA.** Data Access
- **AE.** Alarms and Events
- **HDA.** Historical Data Access
- **VDC.** Volts Direct Current

- **VAC.** Voltage in Alternating Current
- **PAC.** Process Automation Controller
- **CWM.** ControlWave Micro
- **CWRED.** ControlWave Redundant
- **KVM.** Keyboard, Video and Mouse
- **UPS.** Uninterruptible Power Supply
- **W.** Width
- **H.** Height
- **D.** Depth
- **POU.** Program Organization Unit
- **FAT.** Factory Acceptance Test
- **SAT.** Site Acceptance Test

Introduction

Researchers have spent, and are still spending, a lot of time looking for new energy sources. Despite their efforts and discoveries, oil and gas are still the main energy source in the world.

Since 1858, when the first oil well was drilled in Canada, production rates have evolved enormously from 20 barrels a day to over 12 million barres per day in the United States in 2019 [**1**].

Transportation methods and scales had to evolve to respond to the needs of such production rates. One of the common oil and gas transportation methods is pipelines. This method has been used for thousands of years to feed cities with water. It has been adopted in the oil and gas industry for its safety and efficiency. Pipelines also produce less emissions than ships, trucks, and trains [**2**].

Although pipelines are considered safer than ground transportation, failures, errors, and natural disasters can result in pipeline disasters. Poorly maintained pipelines risk fluid and gas leaks. These fluids leak directly into the soil, exposing communities to poisoned ground and surface water along with fumes in the atmosphere. An example of such a disaster is the leak of 33000 gallons of crude oil in Salt Lake City, Utah in 2010 due to a ruptured pipeline. This leak exposed people in a nearby city to chemical fumes, resulting in serious cases of drowsiness and lethargy [**3**]. Gas leaks can cause explosions due to the high volatility of hydrocarbon. An example of such an event is an explosion in San Bruno, California that resulted in destroying over 30 homes, the death of 8 people and the injury of 58 others [**3**].

Therefore, pipeline safety regulations are becoming strict. Thus, the need of safety features and real time monitoring and control capabilities have become a requirement in every oil site.

One of the clients of "Integration Objects" has such a system. It allows him to control his oil and gas sites remotely and monitor the pipelines' states in real time. However, his system has become obsolete, and so, unreliable.

In this context, this graduation project focuses on upgrading the telemetry system of the client's oil site.

This report has three chapters. The first chapter is the project's scope. It contains an overview of the host company, some oil and gas general information, automation related definitions, the project's scope, and client requirements, as well as the project planning.

The second chapter dives into the design of the hardware for the solution. It begins with the layout of the different cabinets of the project. It is where the input and output list is defined as well as the different equipment needed for the project such as controllers and input/output modules. It is also where power requirement of each cabinet is defined, and protection elements are selected. Then wiring diagrams are presented in the power distribution section. Finally, the chapter ends with the arrangement and heat dissipation of each cabinet.

The third chapter, and the last one, presents the software design of the solution. First comes the remote telemetry units' programming. Where the programming software is defined, input/output modules are configured, and controller logic is programmed. Then comes the master telemetry unit's communication protocols configuration. And next are the supervisory control and data acquisition (SCADA) and the human machine interfaces design and configuration. Finally, tests are performed on the solution.

1

General Scope

Introduction

In this chapter the host company ”Integration Objects” will be presented, then an introduction of the Oil and Gas, and the automation industries will be given. After that the project scope and customer requirements will be defined, to finally finish with a planification and the project's GANTT chart.

1 Integration Objects

Founded in 2002 by Mr. Sami Achour, Integration Objects, or IO, is one of the worldwide leading companies in process automation, command and control systems and operational intelligence (logo presented in figure 1.1). Being a member of the OPC Foundation, it provides OPC based decision management solutions to manufacturers in different fileds such as electrical, chemical and Oil and Gas.



Figure 1.1: Integration Objects logo

The company has over 80 personnel divided into three teams:

- **Automation Unit:** whose mission is to design, integrate functional testing, and implement and maintain control and security systems.
- **Developing Unit:** Its mission is to design and develop software products for real-time decision-makig process, and open platform communications connectivity.
- **Advanced Solutions Unit:** Their mission is to design and configure advanced solutions for real-time decision-makig process, for manufacturing performance optimization.

IO's partnership program was developed for distributors, system integrators, independent consultants, value added resellers and original equipment manufacturers such as: OSIsoft, EMERSON, Rockwell Automation, OPC Foundation and SIEMENS. Figure 1.2 shows their logos:



Figure 1.2: Integration Objects partners

Among the solutions provided by Integration Objects:

- Decision support systems

- Emergency response plans
- Management of company's performance and installation
- Telemetry and instrumentation
- Integration of systems

The company operates in different fields such as:
Refining and petrochemical sector, chemical and pharmaceutical industries, the food sector, mining and mineral industries, equipment manufacturers and the Oil and Gas sector.

2 Oil and Gas industry

The oil and gas are major industries in the energy market, and one of the largest sectors in the world in terms of dollar value.
The oil and gas industry can be broken down into three segments [**4**] :

- **Upstream activities** for oil and gas exploration and production. They include discovering deposits, drilling wells and rig operations.
- **Downstream activities** for transforming crude oil into usable products such as gasoline. It also includes the distribution of finished pruducts to end users and retailers.
- **Midstream activities** they link the upstream and downstream entities by transporting and storing oil and gas.

2.1 Oil and gas transportation

Oil and gas, being in its crude state or as a finished product, can be transported via different means depending on the amount and distance:

- Marine vessels or tankers
- Railroad cars
- Trucks
- Pipelines

Since some of these means risk corrosion, the use of cathodic protection is necessary. It is a corrosion control method for protection of metallic structures such as oil and gas pipelines and tankers. It is a method that uses a sacrificial metal to mitigate corrosion of the structure.
There are tow methods for cathodic protection:

- **Sacrificial anode:** it uses a more active metal such as zinc, magnesium and aluminium as anodes connected electrically. The metal provides electrons to the cathode and keeps it from corroding (figure 1.3).
- **Impressed current:** it uses the same principle but with the intermediate of a DC power source to provide current flow between the two metals (figure 1.4).

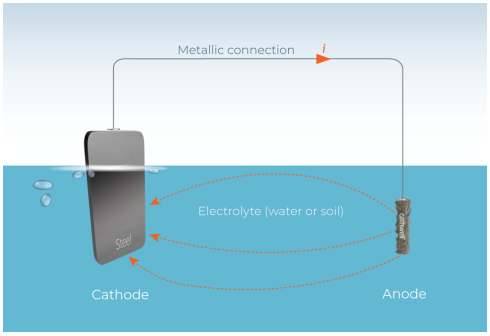


Figure 1.3: Sacrificial anode

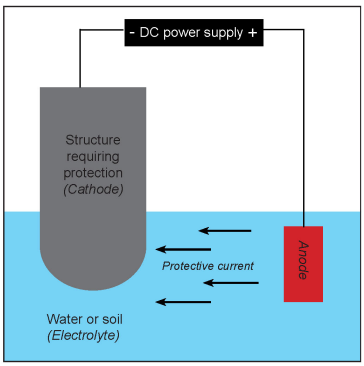


Figure 1.4: Impressed current

2.2 ATEX directives

In oil and gas industries there is a high risk of fire or explosion incidents therefore they are considered a potentially explosive atmosphere.
According to the European commission [**5**], a potentially explosive atmosphere exists

when a mixture of air gases, vapor, mists, or dusts combine in a way that can ignite under certain circumstances.
ATEX directives cover the use of equipment and protective systems intended for use in such atmospheres. They define the essential health and safety requirements and conformity assessment procedures, to be applied before products are placed on the European Union market.

There are two directives:

- **ATEX 137 "workplace" directive 1999:** for minimum requirements for improving the safety and health protection of workers potentially at risk from explosive atmospheres.
- **ATEX 114 "equipment" directive 2014:** for equipment and protective systems intended for use in potentially explosive atmospheres.

3 Automation industry

3.1 Industrial control systems

Industrial control systems or ICS is a broad class of command-and-control networks and systems that are used to support all types of industrial processes.
These systems have evolved from mechanical pneumatic systems to analog controls to today's digital systems.
There are several architectures of ICS the most common of which are SCADA and DCS based.

3.1.1 Supervisory control and data acquisition

Supervisory control and data acquisition, or SCADA, systems are designed to collect field information, transfer it to a central computer facility and display this information to the operator graphically or textually. Allowing the user to monitor and control the entire system in near real time.

The general structure of a SCADA system consists of:

- **Control center:** housing a server that collects information gathered by field sites and displays them on HMIs.
- **Field sites:** contain controllers such as RTU and PLC and filed devices like sensors and actuators.

Communication infrastructures allow the connection between the control center and field sites.

3.1.2 Distributed control systems

DCS or distributed control systems are used to control production systems. They allow the control of various processes individually preventing total system shutdown in case of a failure.
A DCS system generally contains:

- An operator workstation used for monitoring and displaying alarms.
- An engineering workstation used for downloading programs to different controllers, and downloading graphical displays to HMIs.
- Controllers and field devices that supervise individual processes.

The table 1.1 shows the difference between DCS and SCADA systems:

Table 1.1: DCS and SCADA comparison

	DCS	SCADA
Dependency	Vendor dependent	Independent
Communication	Proprietary	Open communications
Software	Integrated HMI/PLC software	Separate software for HMI and PLC
Main orientation	Process oriented	Data gathering oriented

3.2 Controllers

3.2.1 Programmable logic controller

PLC or Programmable Logic Controller is a special form of microprocessor-based controller that uses programmable memories to store instructions and implement functions to control machines and processes (figure 1.5).

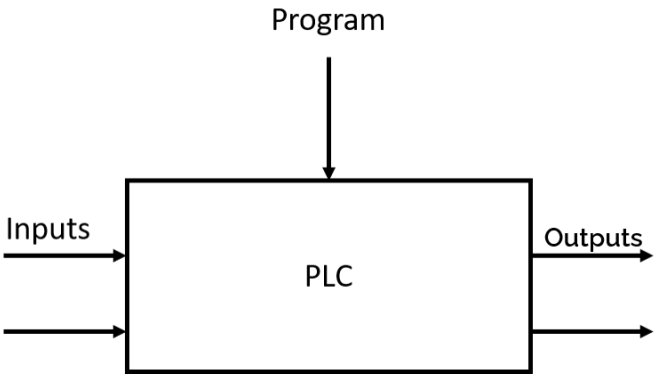


Figure 1.5: Programmable logic controller

The main PLC components are as follows (figure 1.6) :

- **Power supply:** this unit is needed to provide the processor and input/output modules with the necessary voltage.
- **CPU:** interprets input signals and executes control actions, according to the program installed in the memory.
- **I/O modules:** they allow the processor to receive information from external devices and communicate with them.

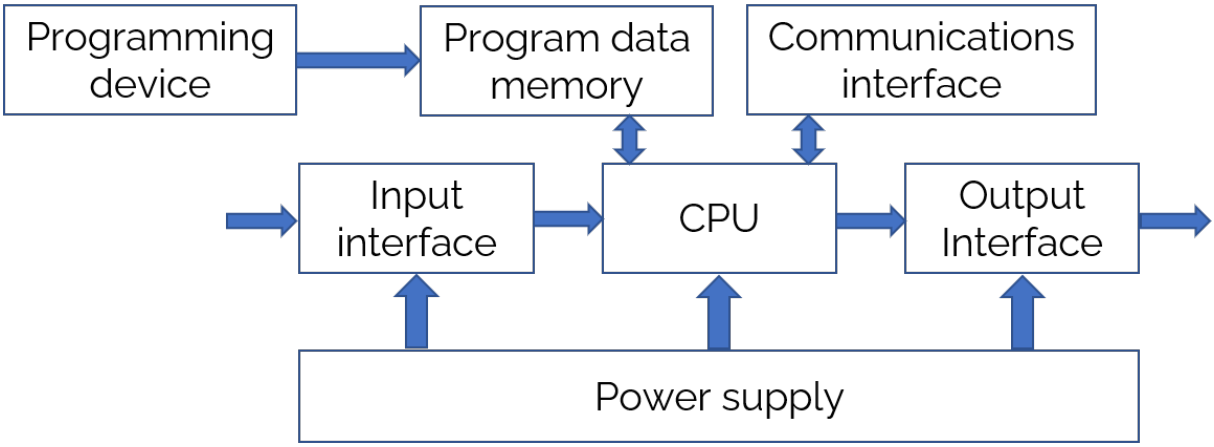


Figure 1.6: PLC system

3.2.2 Remote telemetry unit

RTU or Remote Telemetry Unit was originally developed to provide monitoring and control capabilities at unattended sites. It is used to gather data from field levels and deliver them to a DCS or SCADA system. Besides the same connectivity capabilities of a computer and the input/output capabilities of a PLC that the RTU has, it is also rugged and durable.

Table 1.2 shows some differences between RTU and PLC:

Table 1.2: RTU and PLC comparison

	RTU	PLC
Network scale	Wide telemetry	Local area
Wireless communication	Supported	Not supported
Power consumption	Low power configuration, can run with batteries and solar panel	Need of a permanent power supply
Use	Monitoring	Process control

Some of the known controller manufacturers are presented in figure 1.7.



Figure 1.7: Controller manufacturers

3.2.3 Programming languages

There are several controller programming languages. To standardize programming the IEC (international Electrotechnical Commission) introduced the IEC 611313-3 standard that defines five programming languages:

- Textual: Instruction List (IL) and Structured Text (ST).
- Graphical: Ladder Diagram (LD) and Function Block Diagram (FBD).
- Sequential Function Chart (SFC).

3.3 Industrial communication protocols and standards

Industrial communication protocols are developed to interconnect systems, interfaces and instruments that make up an industrial control system.

3.3.1 Open platform communications standard

Open platform communications, or OPC, is a series of specifications that define the interface between client and server including real time data access, monitoring of alarms and events and access to historical data [**6**].
There are two OPC specifications:

- **OPC Classic** that provides separate definitions for interfaces such as OPC DA, OPC AE and OPC HAD.
- **OPC UA (Unified Architecture)** this architecture unifies all the functionalities of individual OPC Classic specifications into one framework.

3.3.2 Modbus

An open industrial communications protocol in which data transmission is serial. It provides a common language for devices and equipment to communicate with one another. It was created by Modicon, or what is now known as Schneider Electric, in 1979. It has become a standard communication protocol.
Modbus supports master slave network and is placed at the Application (7th) layer of the OSI model. It is a request/reply protocol that offers services specified by function codes.

The OSI model (figure 1.8) is a theoretical model of how communications work on a network and it has 7 layers.

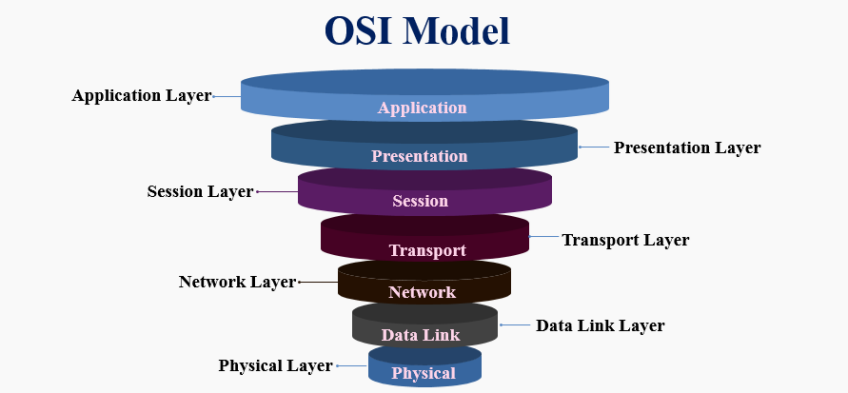


Figure 1.8: OSI model layers

Modbus has several variants two of which are:

- **Modbus RTU:** a very widely used variation that includes error checking.
- **Modbus TCP/IP:** this variation is adapted for use on an Ethernet network.

The modbus RTU frame (figure 1.9) consists of four blocks:

1. **Slave ID:** contains the identification number of the slave

2. **Function code:** this code indicates the function to be executed by the slave to reply to the master's request
3. **Data:** the actual value of the data
4. **CRC:** cyclic redundancy check, an error checking byte

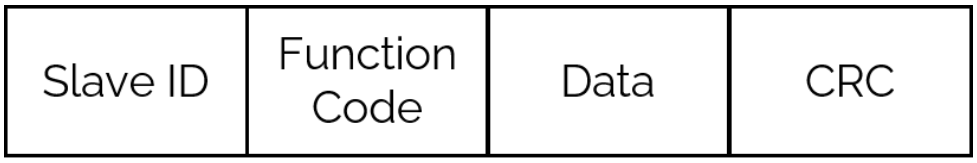


Figure 1.9: Modbus RTU frame

3.3.3 Distributed network protocol 3

DNP3 or Distributed Network Protocol is a non-proprietary protocol that defines communications between master stations, outstations, and other intelligent electronic devices. It was developed to achieve interoperability among systems in different industries.



Figure 1.10: DNP3

Some of the typical DNP3 system architectures are one-on-one (figure 1.11) and multidrop (figure 1.12).

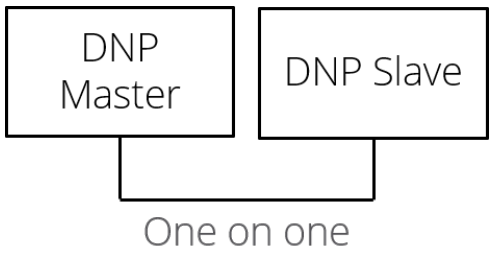


Figure 1.11: One-on-one

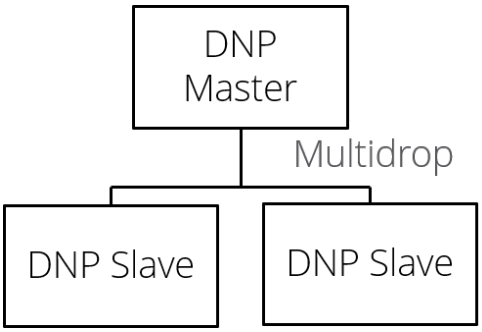


Figure 1.12: Multidrop

4 Project Scope

4.1 Project objective

The objective of this project is to upgrade the telemetry system at an oil site. The existing system has become obsolete and spare parts are becoming scarce and maintenance costs are rising. Therefore, the client wishes to upgrade his oil site's controllers to new equipment with scalability, reliability, and safety as requirements. The project also covers the engineering and design of a new inter-site communication architecture.

4.2 Client requirements

To satisfy the client requirements we need to upgrade three different sites:

- Oil Producer
- Gas Injector
- Manifold

These sites are controlled with Bristol 3305 RTU and an MDS 4710A radio modem. They communicate with a Bristol 3330 MTU via an MDS 4790 radio modem.
The client requires the following upgrades stated in table 1.3

Table 1.3: Equipment upgrade requirements

Equipment	Actual Version	Upgrade to Version
RTU	Bristol DPC 3305	Emerson ControWave Micro
MTU	Bristol DPC 3330	Emerson ControlWave Pac redundant
Radio modem	MDS 4710A	MDS SD4
Radio master	MDS 4790	MDS Master Station

4.3 Input and output documents

Input documents:

- Existing system wiring diagrams
- Actual controller program
- Environmental conditions
- Material specification

Output documents:

- I/O list
- Bill of Materials
- Power consumption calculation
- Heat dissipation calculation
- Electrical diagrams: power distribution, cabinet layout and system's general arrangement
- Programming

4.4 Project planification

The Gantt chart presented in figure 1.13 shows the different tasks executed during the project.

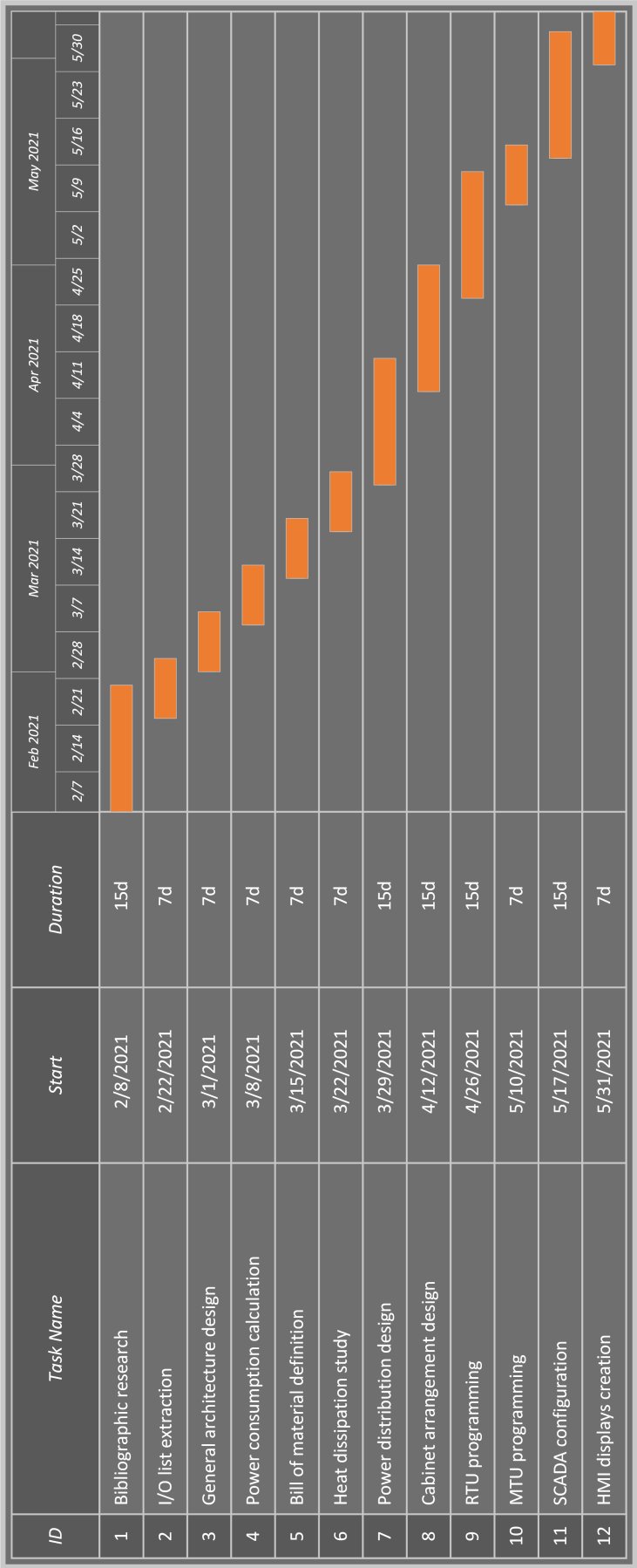


Figure 1.13: Gantt chart

5 Conclusion

In this chapter the general scope of the project was defined by introducing the host company "Integration Objects", setting the client requirements and the project's goal. In addition, the oil and gas industries and some automation related terms were introduced. Finally, the chapter was ended with the planning of the project's different stages and steps. The next step is hardware design and definition.

2

Hardware Design

Introduction

After defining the project scope, the I/O list must be extracted to be able to define the required component and material for the project. Which will allow the calculation of the power consumption and heat dissipation of these components, to finally design the control panels, general arrangement, and the power distribution of the solution.

1 General architecture

The first task that needs to be done is determining the general architecture of the project to be able to proceed. AutoCAD Electrical was used for this task to generate the file presented in the appendix A. the architecture shows a configuration of 3 RTU cabinets connected via radio modems to an MTU cabinet whose role is to collect information from the different sites and share them via ethernet communication to the server cabinet to ensure the monitoring and data acquisition in the control room.

2 Cabinet layout

2.1 I/O list

In order to define the list of inputs and outputs an exhaustive study of the input documents provided by the client was performed. These documents are:

- Existing system wiring diagrams
- Currently used controller programs

The problem with these documents was that there has been changes to the system. These changes were not modified in the mentioned documents. This difference makes defining the I/O list quite complicated.

The first step was extracting the inputs and outputs defined in each document separately, and then these outputs were compared with each other to determine the differences. Finally, after careful consideration the ones that are trivial among them were kept and a final I/O list was generated.

For the first step, the controller program needed a special software to read. It was written in a proprietary language which had to be learned along with the software. For each input/output extracted the following information had to be determined:

- **Signal tag:** this is the identification of the signal
- **Instrument type:** such as pressure transmitters, flow transmitters and switches
- **I/O type:** Analog/ Digital, input/output
- **Signal type:** Dry contact, 4-20mA current loop, 24VDC
- **Range:** this is for analog signals
- **Unit:** For the analog signals

The final step is to add spare signal inputs and outputs for the different sites as a security measure. To do so we add 20% of each signal type per client requirement. Due

to confidentiality matters the extracted I/O list cannot be added to this report. However, the final I/O count as resumed in the table 2.1, is not a problem:

Table 2.1: Final I/O count

Site	AI	Spare	DI	Spare	DO	Spare	Total
Oil Producer	2	3	13	16	7	9	28
Gas Injector	2	3	11	14	5	6	23
Manifold	2	3	12	15	0	0	18

2.2 Equipment list

2.2.1 Input output modules

The choice of the IO modules will be based on the I/O list defined earlier and the ControlWave Micro model configuration manual [**8**] provided by Emerson as shown in table 2.2

Table 2.2: I/O modules selection

System requirements		Chosen Modules
Oil Producer	16 dry contact DI	16 DI 24VDC dry contact module
	9 DO	16 DO open-source output
	3 internally sourced 4-20mA AI	Isolated AI, 8 inputs int sourced 4-20mA
Gas Injector	14 dry contact DI	16 DI 24VDC dry contact module
	6 DO	16 DO open-source output
	3 internally sourced 4-20mA AI	Isolated AI, 8 inputs int sourced 4-20mA
Manifold	15 dry contact DI	16 DI 24VDC dry contact module
	3 internally sourced 4-20mA AI	Isolated AI, 8 inputs int sourced 4-20mA

So, the total needed for the system is:

- 3 x 16DI modules
- 3 x 8AI modules
- 2 x 16DO modules

2.2.2 Controllers

As required by the client the ControlWave Micro controller will be the replacement to its obsolete RTUs and ControlWave Redundant as MTU.

2.2.2.1 ControlWave Micro

Bristol ControlWave Micro from Emerson is a high-performance hybrid controller that combines the capabilities of a PLC, RTU and a PAC. Designed with low power consumption, scalability, and modularity as objectives [**7**].

- **Controller options**

The available ControlWave Micro configurations are as shown in table 2.3 [**8**]

Table 2.3: ControlWave Micro options

Component	Available options	Requirements
CPU	33MHz, 32-bit ARM 9 processor with no Ethernet connection	Clock speed > 12MHz
	150MHz, 32-bit ARM 9 processor with Ethernet connection	Ethernet connection
PSSM	16 to 18 VDC	24VDC power input
	12 to 24 VDC	
Base unit	3 slot panel mount base: PSSM, CPU and one slot for I/O	3 IO modules for each of the Oil Producer and Gas Injector site and 2 IO modules for the Manifold site
	4 slot panel mount base: PSSM, CPU and 2 slots for communication or I/O modules	
	8 slot panel mount base: PSSM, CPU, 2 slots for communication or I/O modules and 2 slots dedicated to I/O modules	

The final controllers configuration choice for each site is then as shown in table 2.4

Table 2.4: Controller configuration

Site	Module	Choice
Manifold	CPU	150 MHz CPU, 2M SRAM, 16M FLASH, 1-ENET & 2-RS 232 & 1-RS 485 (CE Approved)
	PSSM	12-24VDC with key lock & Watchdog with display interface, no wet-end interface (CE Approved)
	Base unit	4 slot panel mount base
Gas Injector	CPU	150 MHz CPU, 2M SRAM, 16M FLASH, 1-ENET & 2-RS 232 & 1-RS 485 (CE Approved)
	PSSM	12-24VDC with key lock & Watchdog with display interface, no wet-end interface (CE Approved)
	Base unit	Isolated AI, 8 inputs int sourced 4-20mA
Oil Producer	CPU	150 MHz CPU, 2M SRAM, 16M FLASH, 1-ENET & 2-RS 232 & 1-RS 485 (CE Approved)
	PSSM	12-24VDC with key lock & Watchdog with display interface, no wet-end interface (CE Approved)
	Base unit	8 slot panel mount base

2.2.2.2 ControlWave Redundant

Emerson ControlWave PAC Redundant controller is a high-performance robust process automation controller that provides exceptional networking capabilities. The redundant controller prevents system shutdowns caused by single point failures. It has dual CPU and power modules and provides optional I/O redundancy [**9**].

The CWRED controller has two CPU options [**10**]:

- 100MHz, 2 RS-232 and 1 Ethernet
- 100MHz, 2 RS-232, 2 RS-485 and 3 Ethernet

The CWRED is powered with dual 24VDC power modules.
For this project, the ControlWave Redundant is required to have at least 2 ethernet ports, hence the choice of the 100MHz, 2 RS-232, 2-RS485 and 3 Ethernet CPU.

2.2.3 Server

The server in this project is required to be able to run the CIMPLICITY V11 software by GE for SCADA and HMI applications. The minimal system requirements for this software are presented in table 2.5 **[11]**

Table 2.5: CIMPLICITY V11 system requirements

Hardware	Requirements
Microprocessor	Intel Core 2 Duo 3.0 GHz
RAM	4 GB
Hard disk	40 GB
Monitor	24-bit graphics card capable of 800 x 600 resolution
Operating system	Windows 10 Professional, Enterprise and Ultimate Editions, windows server 2019

The server chosen is the PowerEdge R540 Rack server from DELL (figure 2.1) with the following specifications:

- **Processor:** Intel Xeon Gold 5222 3.8GHz
- **RAM:** 8GB
- **Storage:** 1TB SSD
- **Video specifications:** : Matrox G200Ew3 24-bit graphics card with 16MB capacity and up to 1920x1200 resolution
- **Operating system:** Windows server 2019



Figure 2.1: PowerEdge R540 server

The use of two R540 servers in the cabinet is essential to ensure the availability of the system. These servers will be linked to a NetController 16 port KVM console from TRIPP LITE (figure 2.2). The KVM will allow the interaction with the servers easily from the cabinet.



Figure 2.2: KVM

2.2.4 Control room

For the control room the equipment in table 2.6 are needed:

Table 2.6: Control room equipment

Equipment	Requirement	Choice
Screen	Resolution > 800x600	Dell 24" S2421HSX
Operator workstation	Cimplicity v11 system requirements	Dell OpTIPLEX 3080 Small Form Factor: 10th generation core i3-101000, 8GB DDR4 RAM, 500GB HDD
Maintenance laptop	Cimplicity v11 system requirements	HP ProBook 455 G8 Notebook, AMD Ryzen 3 5400U, 8GB RAM DDR4, 512GB SSD

2.3 Power consumption

2.3.1 RTU panel

The RTU panel houses the ControlWave Micro controller and the MDS SD4 radio modem.

These panels receive power from a solar panel system delivering 24VDC. Table 2.7 resumes the power needed for each component of the panels according to their technical specification manual provided by the manufacturer [**12**] [**13**].

Table 2.7: RTU panel power consumption

Site	24VDC/ 230VAC	Name	Quantity	Current (A)	Total (A)	Spare current (A)
Manifold	24VDC	CPU, backplane and PSSM	1	0.05	2.654	4
		Local termination I/O module isolated DI	1	0.104		
		Local termination I/O module isolated AI	1	0.3		
		MDS SD4	1	2.2		
Gas Injector	24VDC	CPU, backplane and PSSM	1	0.05	2.81	4
		Local termination I/O module isolated DI	1	0.104		
		Local termination I/O module isolated AI	1	0.3		
		Local termination I/O module isolated DO	1	0.012		
		Relay module	16	0.144		
		MDS SD4	1	2.2		
Oil Producer	24VDC	CPU, backplane and PSSM	1	0.05	2.81	4
		Local termination I/O module isolated DI	1	0.104		
		Local termination I/O module isolated AI	1	0.3		
		Local termination I/O module isolated DO	1	0.012		
		Relay module	16	0.144		
		MDS SD4	1	2.2		

2.3.1.1 Protection elements

- **Terminal blocks**

Terminal blocks are used to link two components or more in an electrical circuit. They offer a certain level of protection to the circuit and allow for better cable management and cabinet organization.
The existing terminal blocks will be used in power distribution for the cabinets of the different sites.

- **Lightning protector**

The lightning protector flows the impulse current, during a lightning strike, to earth and returns to its initial state. It is a safety precaution taken by the client in the existing cabinets and therefore will be kept in the new solution.

- **Circuit breaker**

Circuit breakers are essential in every electrical circuit as they provide protection against short circuits and unwanted current spikes. According to the current requirement in table 2.7, RTU cabinets need a total of 4A of current and based on the available circuit breaker specifications available in the market the 5A is the closest. And so, the choice of a 5A 2 pole C curve circuit breaker from ROCKWELL AUTOMATION (figure 2.3) was made.
C curves are intended for use in circuits with medium induction loads such as control panels. The circuit breakers used in this project are of C class.



Figure 2.3: 5A circuit breaker

- **Fused terminal blocks**

Fused terminal blocks are similar to the feed through terminal blocks, but they offer an extra layer of protection with an integrated fuse. The old system had fused terminal blocks that will be used in the new system with different fuse ratings as shown in table 2.8

Table 2.8: CPU and PSSM requirement

Fuse	Current rating
MDS SD4	3
PSSM	0.062
DO module TB1 pin 9	0.4
DO module TB2 pin 9	0.4
Output Relays	0.5

Fuse ratings selected previously are standard values available in the market which makes them easy to get.

2.3.2 MTU panel

As for the MTU panel, the power received comes from a 230VAC 50Hz power source, which needs to be converted to a 24VDC signal for the CWRED and MDS master station.
This panel's power consumption is presented in table 2.9 **[14]**

Table 2.9: MTU panel power consumption

Cabinet	24VDC/ 230VAC	Name	Quantity	Current (A)	Total (A)	Spare current (A)
MTU cabinet	230VAC	ControlWave PAC Redundant PSSM and CPU	2	0.6	5.6	7
		CCRSM	1			
		MDS Master station	1	5		

2.3.2.1 Power converter

The 230VDC power signal provided by the source needs to be adapted to the needs of the ControlWave Redundant and the MDS master station which use a 24VDC signal for power. That is where an AC/DC converter is needed. Two UNO2 230VAC/24VDC 10A AC/DC converters from PHOENIX CONTACT (figure 2.4) are used in redundancy to provide the system with the required power. These units have the following specifications:

- **Input voltage:** 230VAC
- **Output voltage:** 24VDC
- **Output current:** 10A



Figure 2.4: AC/DC converter

This module consumes 1.2A of current (2.4A for both).

2.3.2.2 Protection elements

- **Circuit breaker**

As this cabinet has redundant power supply, 2 circuit breakers will be needed. Since the power converters require 1.2A of current, two 1.6A C curve circuit breakers from ROCKWELL AUTOMATION (figure 2.5) will be used.



Figure 2.5: 1.6A circuit breaker

Since the MDS Master station requires 5A of current it is best to use a circuit breaker as protection, therefore a 6A breaker from ROCKWELL AUTOMATION is used.

- **Fused terminal blocks**

Since information about the existing MTU cabinet was not provided, it was decided to deal with it as if it does not exist. So, the fused terminal blocks in this cabinet will be provided by PHOENIX CONTACT.
Fuses for these terminal blocks are rated at 0.6A

- **Terminal blocks**

Same as for the fused terminal blocks, these feed through terminal blocks will be provided by PHOENIX CONTACT.

- **Diode module**

The diode module serves for a redundant power input to the MTU cabinet. As power is coming from the mains grid and from a UPS, the diode module allows for a seamless

switching of power sources in case of a failure. The module chosen is a 10A 24VDC module from PHOENIX CONTACT (figure 2.6) .



Figure 2.6: Diode module

2.3.3 Server cabinet

The server cabinet is also powered by a 230VAC 50Hz power source. With 20% spare the total current requirement for this cabinet is 20A as shown in table 2.10. For this cabinet, two 20A circuit breakers from ROCKWELL AUTOMATION are needed as protection for the circuit.

Table 2.10: Server panel power consumption

Cabinet	24VDC/ 230VAC	Name	Quantity	Current (A)	Total (A)	Spare current (A)
Server cabinet	230VAC	Server	2	4.5	16.6	20
		KVM	1	0.8		
		Switch	2	3.4		

3 Power distribution

Power distribution diagrams are documents used to define the power distribution inside the cabinet for implementation and maintenance purposes. These documents are designed using AutoCAD Electrical, as it has a wide library of electrical components and allows the creation of new components.

Each document contains a power distribution diagram along with the different input and output terminations as shown in the Appendix B.

IO terminations define the wiring of the different devices to the controller's modules and jumper configuration if needed.

The figure 2.7 shows the Digital Inputs termination of the gas injector site.

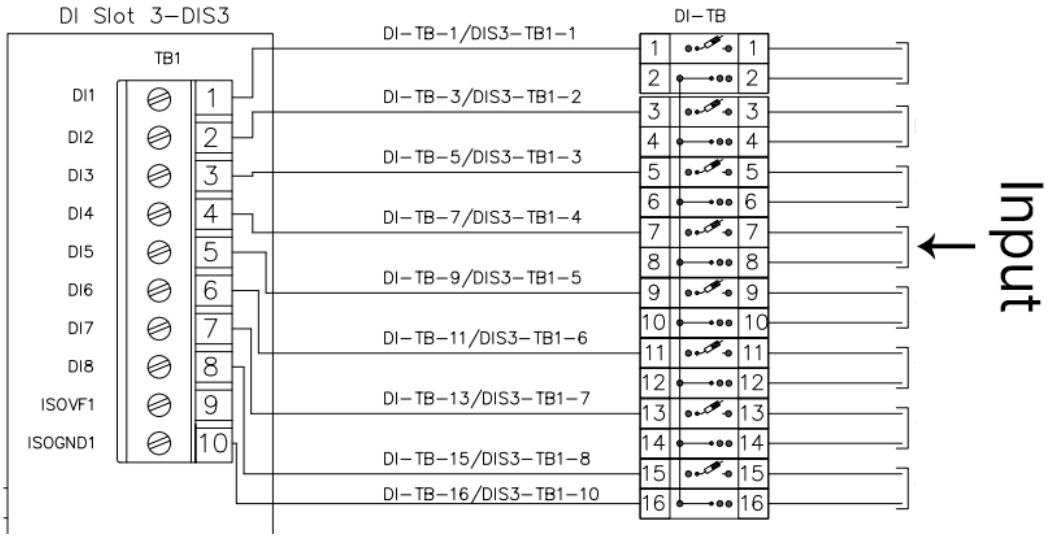


Figure 2.7: DI termination

Figure 2.8 shows the jumpers configuration of the digital input module.

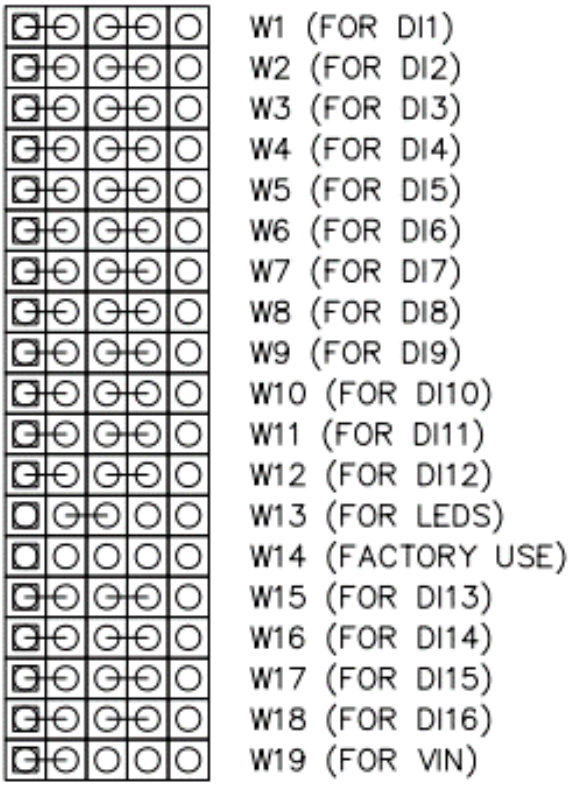


Figure 2.8: DI Jumpers configuration

4 Cabinet arrangement

These documents show the cabinet layout. AutoCAD Electrical was used for these documents as it has the capability of electrical panel creation.

4.1 Manifold cabinet

For the manifold site, the existing cabinet (figure 2.9) will be used as its size is big enough to house the components listed in the section above. It is a flameproof stainless steel enclosure suitable for this application.

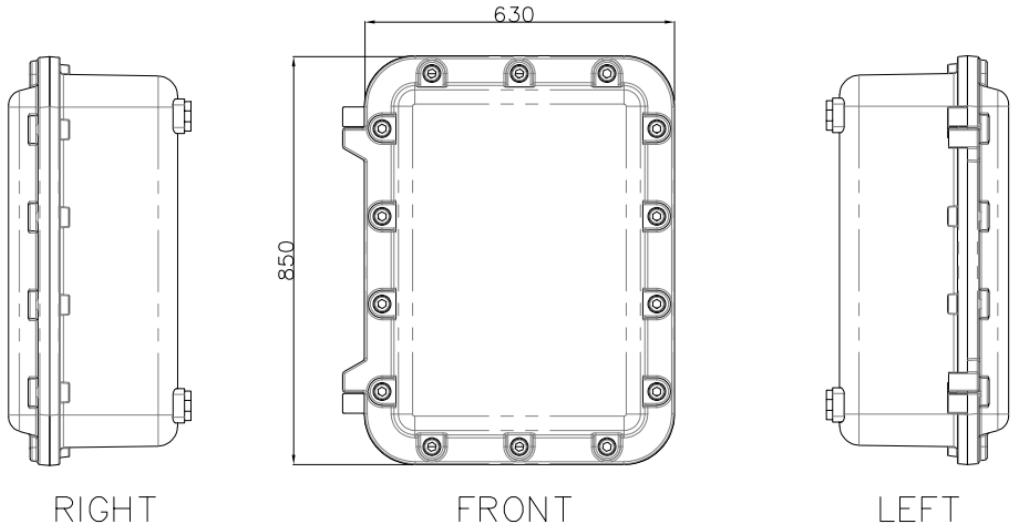


Figure 2.9: RTU cabinet views

Din rails will be needed to mount the circuit breakers, lightning protector and terminal blocks to the cabinet. The ControlWave Micro and MDS SD4 are mounted directly to the cabinet's back plate.

4.2 Gas Injector cabinet

The existing gas injector cabinet will also be used for this site, it is the same as the one used for the manifold cabinet in size, shape, and material. The difference in the arrangement is due to the addition of the relay modules for the digital outputs which makes it a tight fit as shown in the appendix C.

4.3 Oil Producer cabinet

This cabinet is the same as the gas injector cabinet in terms of the actual cabinet and the arrangement.

4.4 MTU cabinet

The MTU cabinet (figure 2.10) was chosen according to 3 criteria as shown in table 2.11

Table 2.11: MTU cabinet choice criteria

Criteria	Requirement	multicolumn1c| Choice
Size	WxDxH > 483mmx406mmx400mm	WxDxH = 600mmx600mmx700mm
Mounting option	1Rack mount for the CWRED	14U rack mount enclosure
Flame resistance	Flame resistant	For use in zones 1,2 and 21 and 22

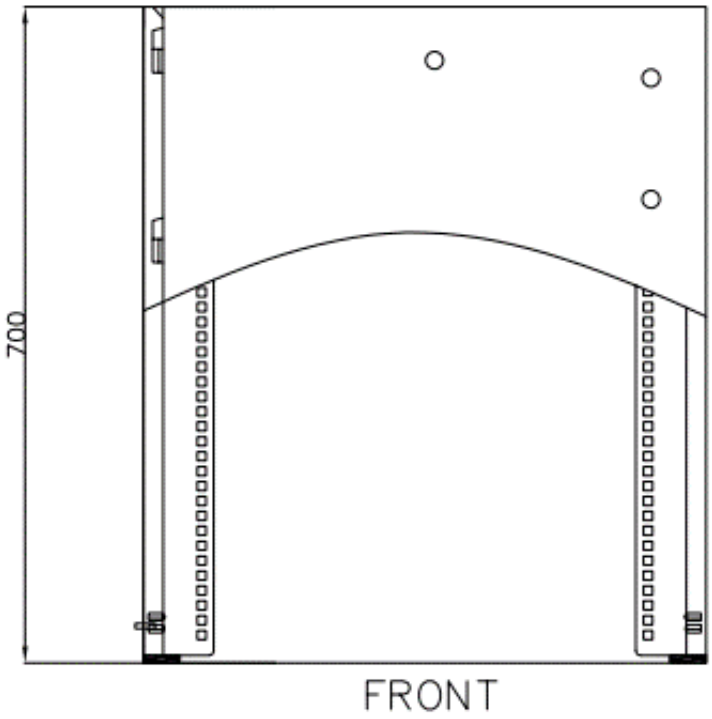


Figure 2.10: MTU cabinet front view

The full cabinet arrangement is shown in appendix D.

4.5 Server cabinet

For the server cabinet the biggest constraint is the depth since the PowerEdge R540 is almost 700mm deep. With that dimension the choice was quite limited and adding the wall mount requirement made it almost nonexistent. However, TRIPP LITE has a wall mount server depth enclosure (figure 2.11) that fits the requirements.

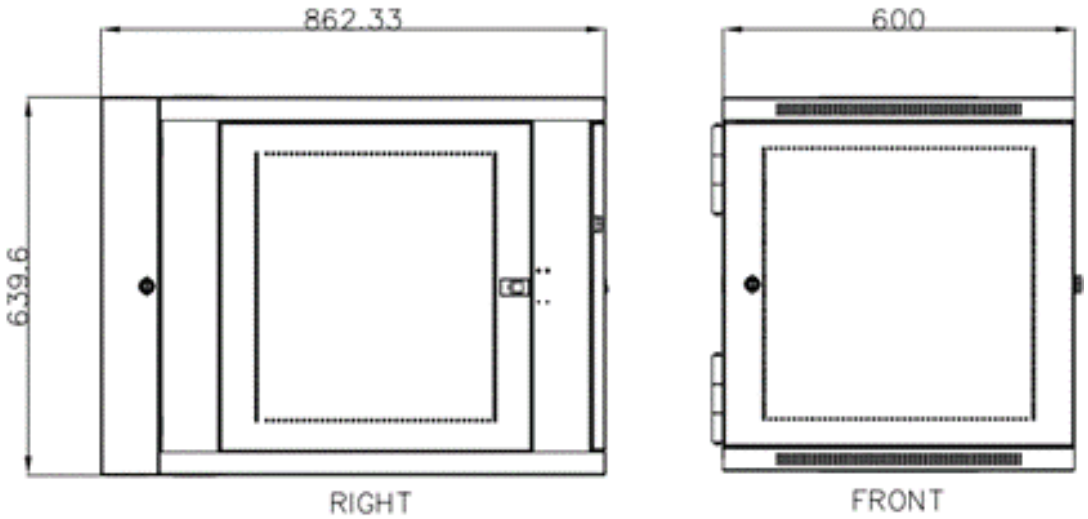


Figure 2.11: Server cabinet front and side views

This cabinet allows access from the front as well as from the side with its removable side plates. It is a 12U rack enclosure constructed from heavy duty steel.

On the right side of the cabinet electrical sockets are mounted on a din rail, these sockets deliver power to the servers, KVM, and the fans mounted at the top and bottom of the cabinet. On the left side are mounted the different circuit breakers and terminal blocks. The appendix E shows the arrangement in detail.

5 Ventilation

Every electrical component dissipates heat while it is turned on, therefore a heat dissipation study was performed to determine which cabinets needed ventilation to keep the temperature inside low. This study has three steps:

- Determining the total heat dissipated by the equipment in each panel Qa (W)
- Determining the heat dissipated by the enclosure according to the formula:

$$Qe = h * A * \Delta T$$

Where:

Qe: heat exchange across the panel's surface (W)

h: heat transfer coefficient (W/m^2K)

A: surface of the enclosure (m^2) (formula in figure 2.12)

ΔT: temperature difference between the interior and exterior of the panel

Enclosure installation type according to IEC 60 890	
	Single enclosure, free-standing on all sides A = 1.8 · H · (W + D) + 1.4 · W · D
	Single enclosure for wall mounting A = 1.4 · W · (H + D) + 1.8 · H · D
	First or last enclosure in a suite, free-standing A = 1.4 · D · (W + H) + 1.8 · W · H
	First or last enclosure in a suite, for wall mounting A = 1.4 · H · (W + D) + 1.4 · W · D
	Enclosure within a suite, free-standing A = 1.8 · W · H + 1.4 · W · D + H · D
	Enclosure within a suite, for wall mounting A = 1.4 · W · (H + D) + H · D
	Enclosure within a suite, for wall mounting, covered roof surfaces A = 1.4 · W · H + 0.7 · W · D + H · D

Figure 2.12: Useful surface determining formula

- Determining the heat that needs to be ventilated Qv (W)

$$Qv = Qe - Qa$$

The table 2.12 resumes the calculation for the different cabinets

Table 2.12: Ventilation study

Site /Cabinet	W(m)	H(m)	D(m)	A(m^2)	ΔT	Qe(W)	Qv(W)	Ventilation needed
Manifold	0.36	0.85	0.35	1.59	10	71.72	55.80	NO
Gas Injector	0.36	0.85	0.35	1.59	10	71.72	50.80	NO
Oil Producer	0.36	0.85	0.35	1.59	10	71.72	50.80	NO
MTU cabinet	0.6	0.7	0.6	1.84	10	83.16	10.16	NO
Server cabinet	0.6	0.635	850	2.22	15	149.89	-4332.90	YES

So, the only cabinet that needs ventilation is the server cabinet, hence the use of fans at the top of the cabinet.

The fans used are two 230VAC high-performance fans capable of dissipating more than 10kW of heat.

6 Conclusion

In this chapter the system's I/O was defined, which allowed the listing and sizing of the hardware needed for the system, which led to the configuration of power distribution and design of the different cabinets and their layout. Now the remaining task is the software design

3

Software Design

Introduction

This chapter will cover the control logic of the controllers, the SCADA configuration and HMI displays, and the communication setup between RTU-MTU and MTU-SCADA.

1 RTU programming

1.1 ControlWave designer

ControlWave Designer is included in the OpenBSI utilities pack, it is used to implement the control logic of ControlWave controllers such as the CWM. It supports the IEC standard programming languages such as the ST, LD, FBD, IL, and SFC. It can compile and download the logic into the connected controller and has a simulation capability that is useful for testing.

1.2 Input/output modules configuration

Before jumping to programming, the controller's modules need to be defined.
The I/O Configuration Wizard makes this configuration in 3 simple steps:

- **First step:** select from available I/O configurations and resources (figure 3.1). Since most projects use a single configuration and resource, this step is generally skipped when first starting the wizard.

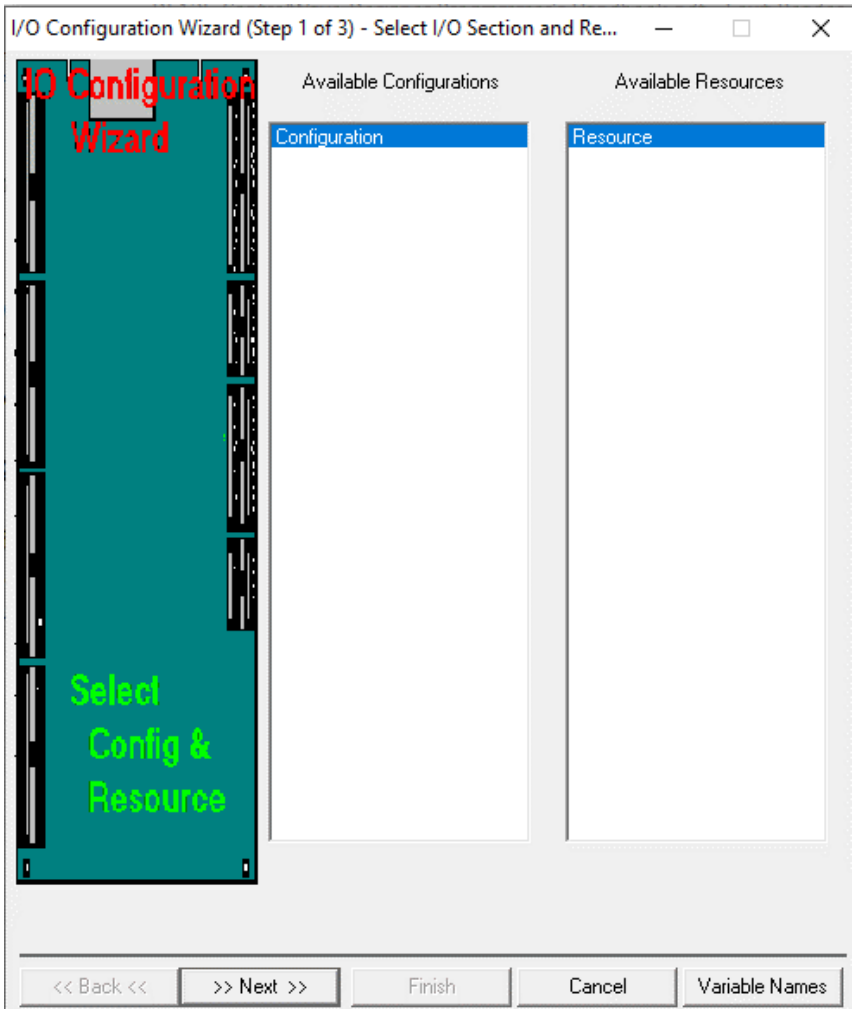


Figure 3.1: Step 1 I/O configuration wizard

- **Second step:** in this step (figure 3.2) the user identifies I/O boards that are installed in the controller or installed separately, such as remote I/O boards.

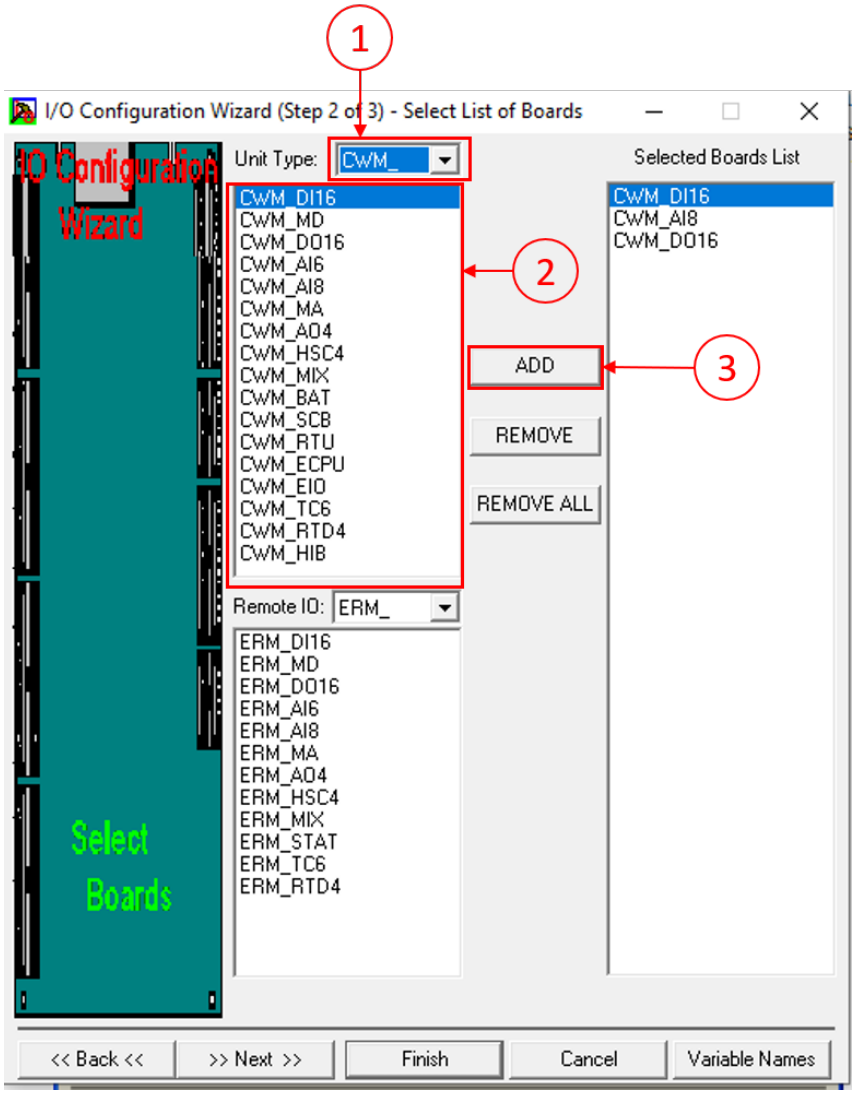


Figure 3.2: Step 2 I/O configuration wizard

1. Select controller type
2. Select board (or remote I/O board)
3. Click ”ADD” to add the board to the selected boards list

Note: boards should be selected in the ascending order of their slot number.

- **Third step:** configure details for each selected board (figure 3.3)

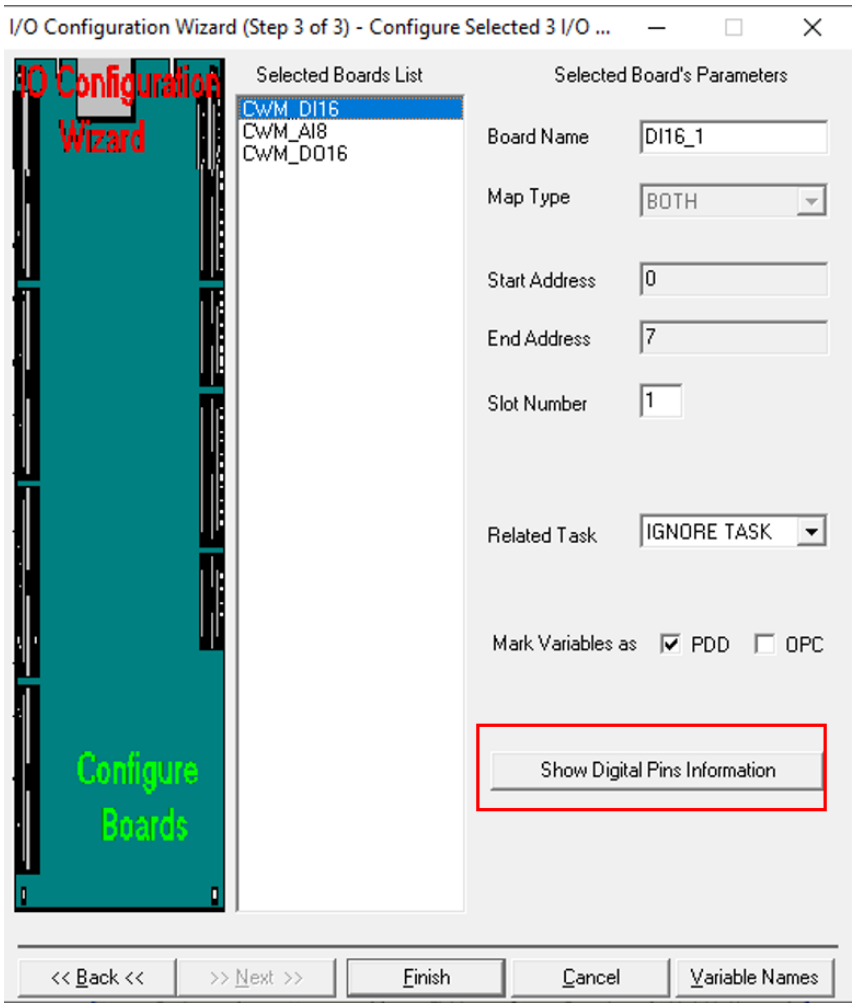


Figure 3.3: Step 3 I/O configuration wizard

To configure the details, the user needs to click on the desired board and then click the ”show digital pins information” button. Note that the text of this button could change depending on the board type.

For the analog input board (figure 3.4), double click the pin number to activate it, type in the name or tag and define the different parameters like span and zero and range type.

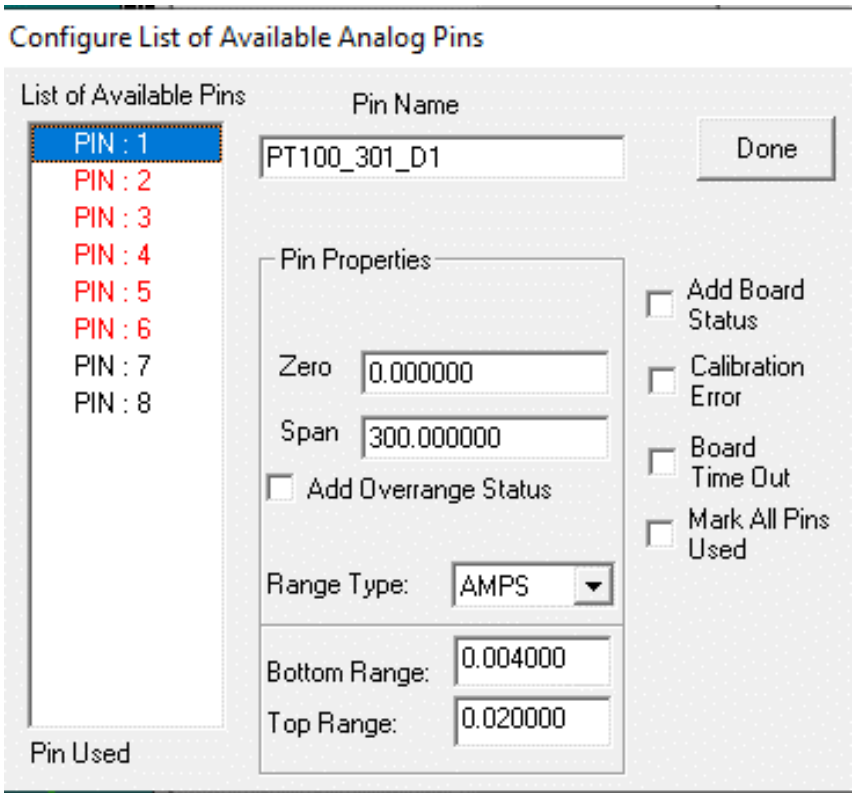


Figure 3.4: Analog pins information window

When done click on ”Done” and then ”Finish” to exit the I/O Configuration Wizard.

1.3 Program

The majority of the RTU logic is configured mainly using the FBD programming language along with some LD and ST. The program can be divided into four sections:

- Valve monitoring
- Pressure transmitter monitoring
- Alarms generation
- Communication configuration

1.3.1 Valve monitoring

This section is responsible for defining the different states of the valves as shown in table 3.1

Table 3.1: Valve monitoring

	Condition	Output
Discrepancy	Closed limit switch XNOR open limit switch	Valve fault
Valve open	Open when open limit switch = 1	Valve open digital output
Valve closed	Closed when closed limit switch = 1	Valve closed digital output

Configuring this logic is a simple task using function blocks as show in figures 3.5 and 3.6

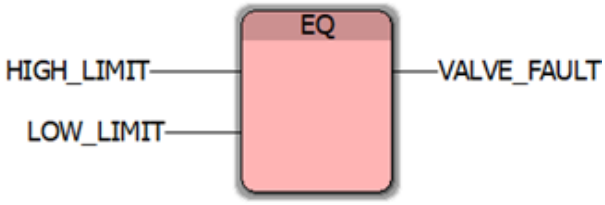


Figure 3.5: Valve discrepancy

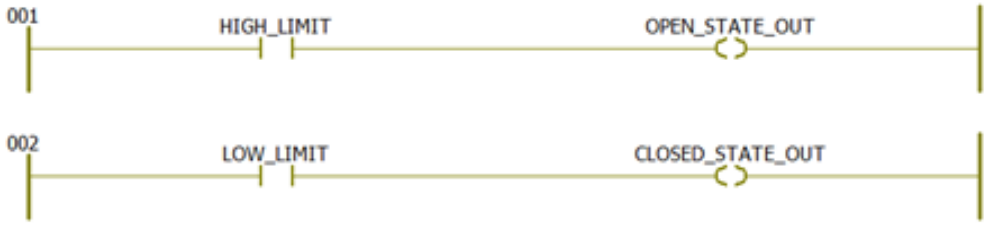


Figure 3.6: SCADA signals definition

HIGH_LIMIT → open limit switch

LOW_LIMIT ⇒ closed limit switch

1.3.2 Pressure transmitter monitoring

This is where the pressure value is monitored and compared to predefined high and low limits as shown in table 3.2.

Table 3.2: Pressure transmitter monitoring

	Condition	Output
Pressure out of range	(Pressure > high range) OR (pressure < low range)	Pressure out of range alarm
Pressure high	Pressure > high limit	Pressure high alarm
Pressure low	Pressure < low limit	Pressure low alarm

Figures 3.7, 3.8 show the program implementation of this configuration in ControlWave Designer

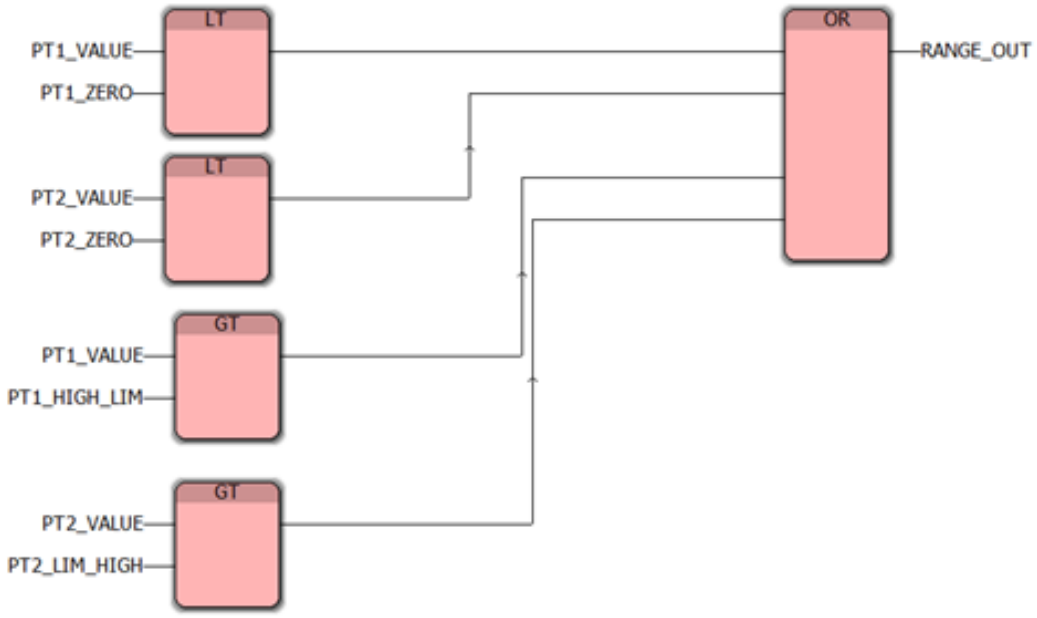


Figure 3.7: Pressure out of range



Figure 3.8: Pressure alarms

1.3.3 Alarms generation

Alarms are generated according to the state of inputs received from field devices and the conditions set in the controller's program as shown in table 3.3.

Table 3.3: Alarms generation conditions

	Condition	Output
Fire detected	Fire detected switch = 1	Fire detected alarm
Solar panel system fault	Solar panel input = 1	Power fault alarm
Control panel and RTU common	RTU control panel input = 1	Common alarm
Battery state	Battery ok = 0	Battery low alarm

The battery ok variable is an internal system input that indicates the State of the RAM battery. 1 state indicates that the battery is working correctly, a 0 state indicates a low battery.

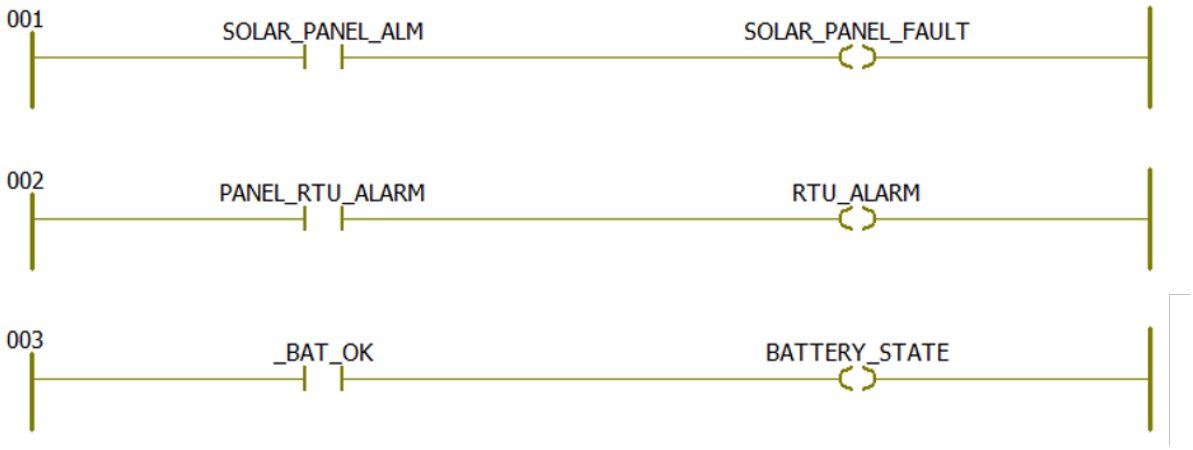


Figure 3.9: RTU alarms



Figure 3.10: Fire alarm

1.3.4 Communication configuration

The communication between the RTU and MTU is through the Modbus communications protocol. The RTU and MDS SD4 radio module are linked by serial communication,

hence the use of Modbus RTU variation of the protocol.

The RTU will play the part of a Modbus slave device which responds to requests sent by the master (in this case the MTU). For this communication to work some parameters need to be configured:

- The serial port used for the communication needs to be set for a Modbus communication with the desired baud rate.
- The device's address (number) in the network
- The communication mode: in this case Modbus RTU slave mode
- Coils list
- Registers list
- Exception status list

The serial port configuration is done with the flash configuration tool of the OpenBSI pack. The rest is configured within the ControlWave Designer software.

The "Custom" function block is the one used for all kinds of communications. For a Modbus slave configuration (figure 3.11) , the inputs that are relevant are the Mode, Customlist, ComPort, and the SlaveAddress. These inputs are configured as shown in table 3.4.

Table 3.4: Slave custom block inputs

Input	Description	Value
iMode	Specifies one of the supported communication interfaces	7
iCustomlist	The number of a parameter list that defines the unique characteristics of the communications interface	1
iComPort	Identifies the serial port used for custom communications	1
iSlaveAddress	Specifies the address of the Modbus slave	1

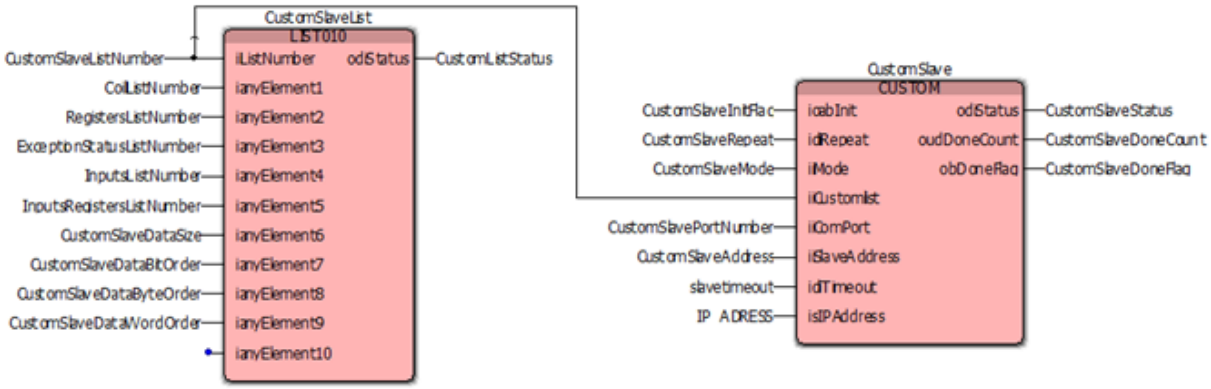


Figure 3.11: Modbus slave configuration

2 MTU programming

The MTU is the unit that will collect data from all the sites and send it to the SCADA system. The communication between this controller and SCADA is through the DNP3 protocol. So, it needs to be configured as a Modbus master and DNP slave.

2.1 Modbus master

The configuration of this communication is basically done in the same way as the slave configuration through the dedicated custom block (figure 3.12) . However, some parameters need to be changed. Table 3.5 shows the values needed for the required inputs of the custom block.

Table 3.5: Modbus master custom block inputs

Input	Value
iMode	4
iCustomlist	1
iComPort	1
iSlaveAddress	1

In the custom list the first parameter specifies the Modbus function code. This code

tells the slave which function to perform. Some of the supported functions by Modbus are presented in table 3.6

Table 3.6: Modbus function codes

Function	Function code
Read coil status	1
Read holding registers	3
Force single coil	5
Preset single register	6

Parameter 2 coil/register number: it specifies the starting address for coil or register operations.
Parameter 3 count: a parameter whose value specifies the number of coils or registers to be read.
Parameter 4 input/output list number: the number of the list used to control input/output data.

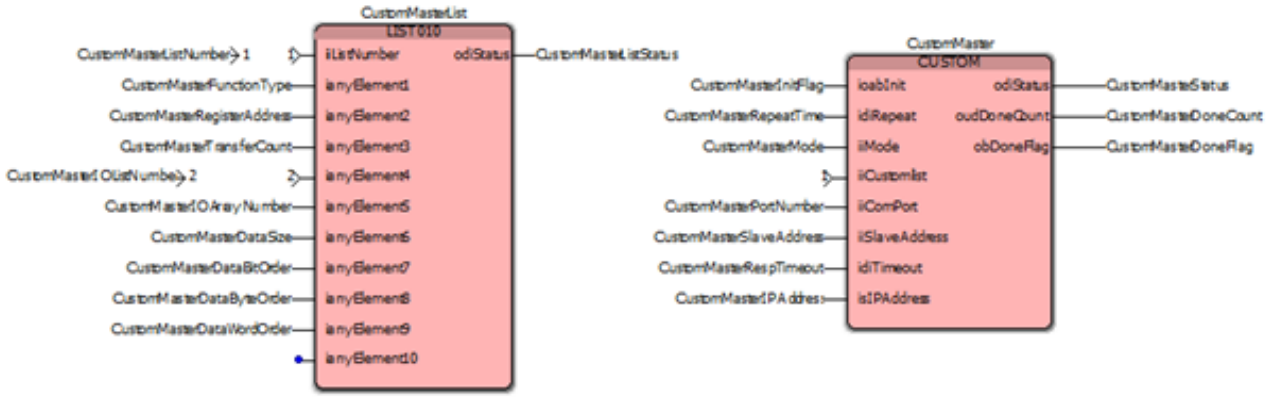


Figure 3.12: Modbus master configuration

2.2 DNP3 outstation

The MTU will be the DNP outstation (slave) which will respond to requests sent by the client (master) in this case SCADA.

To configure a DNP communication some parameters, need to be defined first:

- Channel parameter structures used to define connection points
- Outstation slave status structure
- Outstation sessions to be used
- Outstation valid connections
- Data types such as binary output points and analog output points

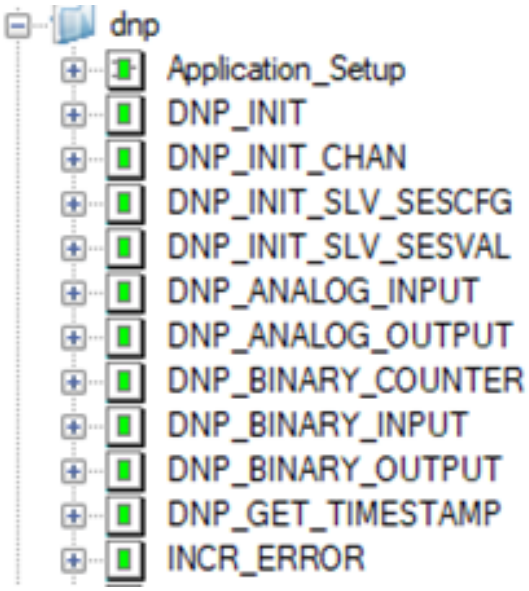


Figure 3.13: DNP configuration POU

3 SCADA and HMI

3.1 Cimplicity

Cimplicity is a software for control and visualization. It is an HMI and SCADA platform developed by GE. Cimplicity v11 allows several controls and configurations from alarm definition and classes to points devices and communication ports in addition to different security parameters such as roles and user definition.

3.2 System configuration

System configuration is necessary to plan for the configuration needed to implement in the SCADA system.

3.2.1 Users

For the project there are different users with different levels of authority and privileges that need to be defined and configured in the system. The users for this project and their roles are as shown in table 3.7

Table 3.7: Users configuration

User	Role	Privileges
Supervisor	SYSMGR (system manager)	• Set points • View alarms • Acknowledge alarms • Delete alarms
Engineer	ENG	• Set points • View alarms • Acknowledge alarms
Operator	OPER	• View alarms • Acknowledge alarms

3.2.2 Alarms

As mentioned before Cimplicity allows the configuration of alarms and their classes and behaviors. The alarms needed for this project are as follows in table 3.8:

Table 3.8: Alarms configuration

Alarm	Class
RTU battery ok	RTU
Solar panel fault	
RTU fault	
Valve Discrepancy	VALVE
Pipe pressure high	HIGH
Pipe pressure low	
Pipe range out	
Fire detected	

RTU class takes care of alarms that are related to the RTU and its systems such as the solar panel system.

The Valve class is for the alarms in relation to the valve's faults and states.

HIGH alarm class is for alarms with high priority such as fire detection.

3.2.3 Points

Points hold data received from slaves or servers and any results calculated within the SCADA system. The system points are defined as shown in table 3.9.

Table 3.9: Points configuration

Object	Point	Type
V30X	Open state	BOOL
	Closed state	BOOL
	Discrepancy	BOOL
PT30X	Value	REAL
	Pressure low alarm	BOOL
	Pressure High alarm	BOOL
	Pressure limit high	REAL
	Pressure limit low	REAL
	Range high	REAL
	Range low	REAL
	Range out	BOOL
	Fire detection	BOOL
	Solar panel fault	BOOL
	RTU fault	BOOL
	PSD command	BOOL
	RTU battery ok	BOOL

3.3 HMI screens

Cimplicity has an integrated HMI designer that allows a seamless and easy displays creation.

This project has 3 main screens, an alarm viewer display and device faceplates.

The first main screen is the Home screen (figure 3.14) in which the user selects a site to monitor. Buttons that have no name represent other wells in the same site that need to be added. During this project, work has only been done on one well for each site.

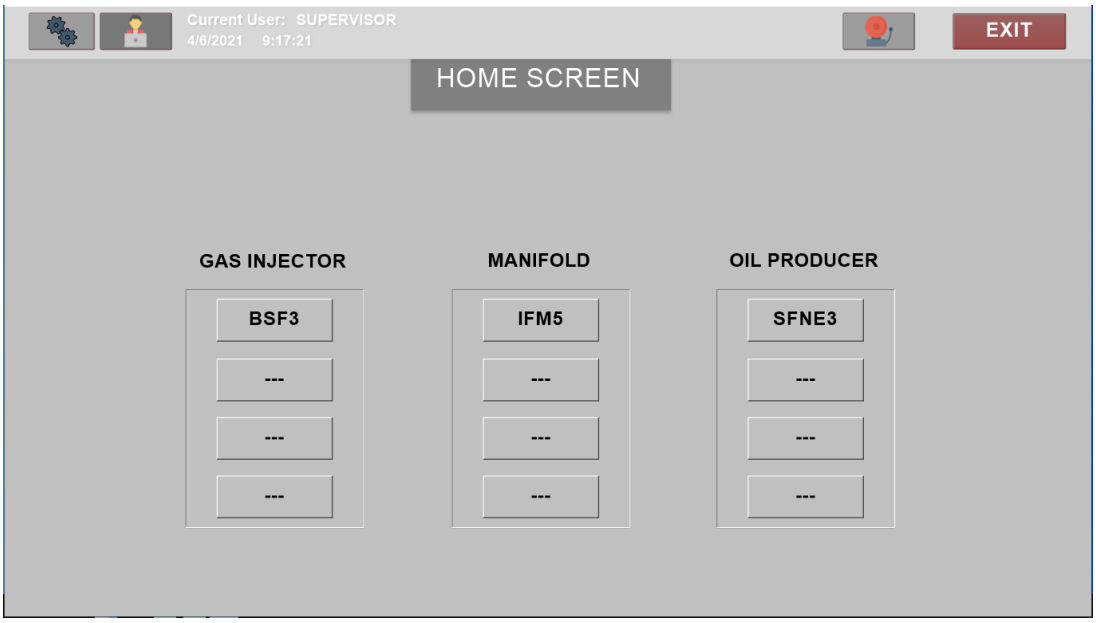


Figure 3.14: Home screen

Once a site has been selected the user is then taken to the site's control screen (figure 3.17). In this screen the user can monitor the state of the valves and the pressure transmitters.

A click on one of the valves opens the device's faceplate (figure 3.15) where the user is able to monitor in detail its state and get the error messages in case of a fault. Similarly, if a pressure transmitter is clicked on, this one's faceplate appears (figure 3.16), allowing for further control and monitoring. In this faceplate the user can read the exact pressure value, the high and low ranges, and the upper and lower limits for alarms. A chart on the left gives further insights on the pressure values for better analysis.

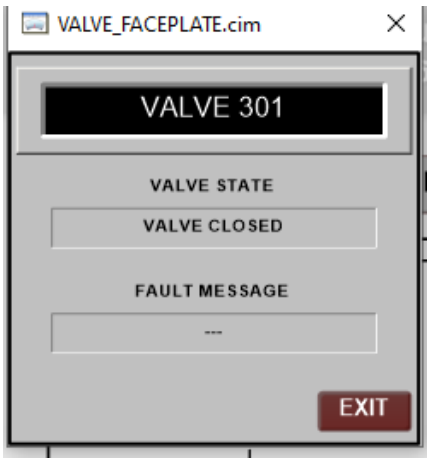


Figure 3.15: Valve faceplate

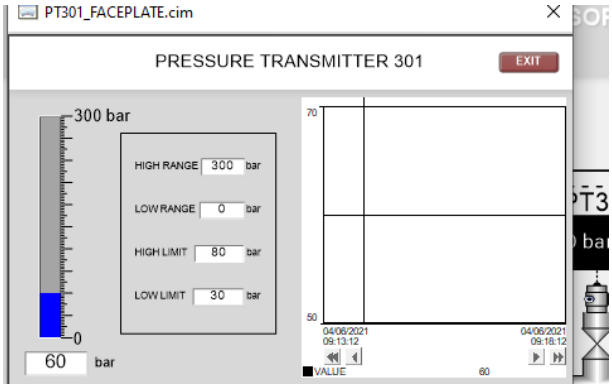


Figure 3.16: Pressure transmitter faceplate

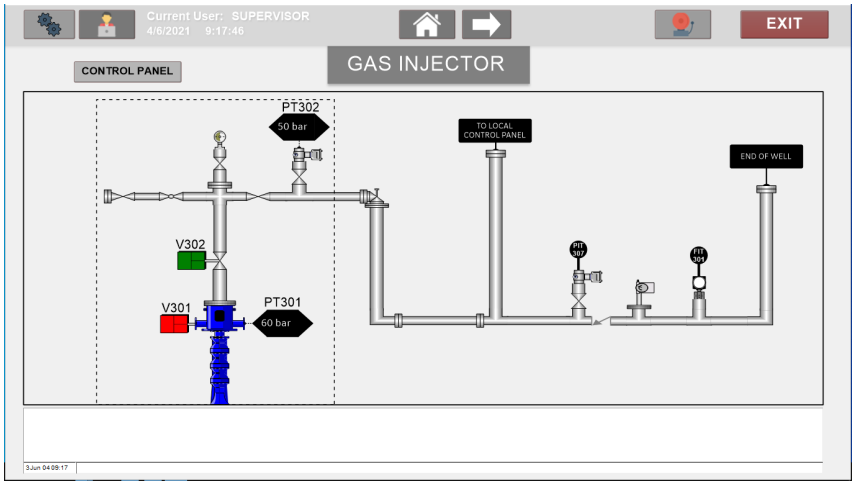


Figure 3.17: Gas Injector screen

On the right corner near the exit button there is a button that takes to the alarm viewer screen where all alarms are displayed (figure 3.18).

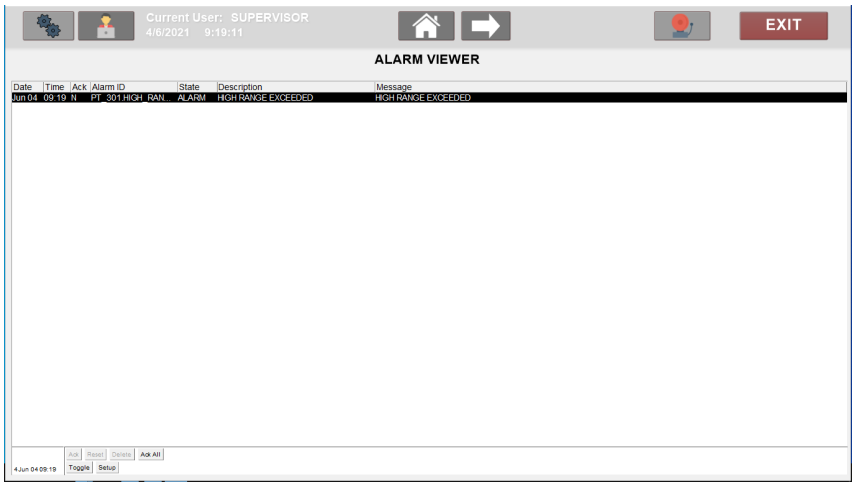


Figure 3.18: Alarm viewer screen

In this screen, depending on its role, the user can acknowledge, reset, and delete alarms.

The user can also go to the diagnostics screen (figure 3.19) in which RTU and MTU states can be accessed (figure 3.20).

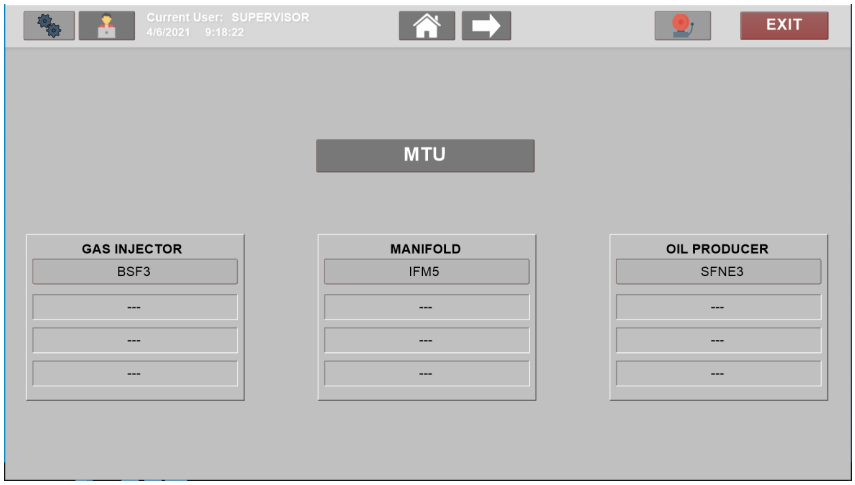


Figure 3.19: diagnostics screen

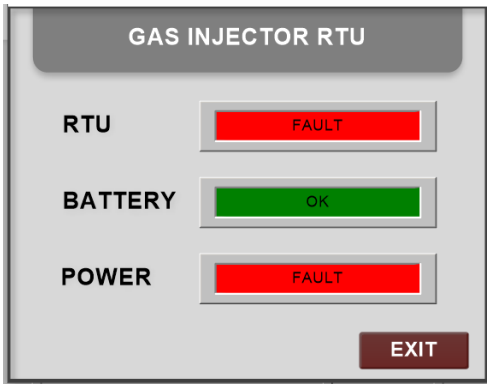


Figure 3.20: RTU faceplate

3.4 DNP3 configuration

Despite having a lot of features and supporting different communication protocols, Cimplicity does not have a built in DNP driver. Therefore, a separate driver needs to be installed to allow such communication.

Once the Catapult DNP3 client driver is installed a DNP3 option appears in the protocol selection tab of the project properties in Cimplicity as shown in figure 3.21.

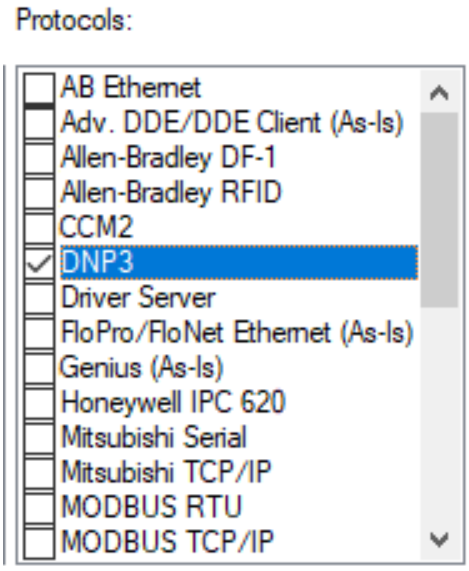


Figure 3.21: DNP3 protocol option

After this protocol is selected, a DNP port and a device can be defined in the project as illustrated in figures 3.22 and 3.23.

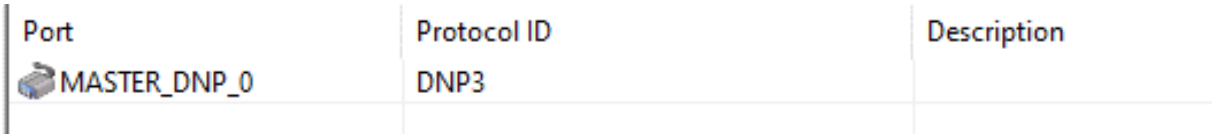


Figure 3.22: DNP port



Figure 3.23: DNP device

From within Cimplicity it is only possible to define a serial DNP communication. However, this does not matter since the driver only cares about the name of the device defined in Cimplicity. The actual definition is done on the driver interface (figure 3.24).

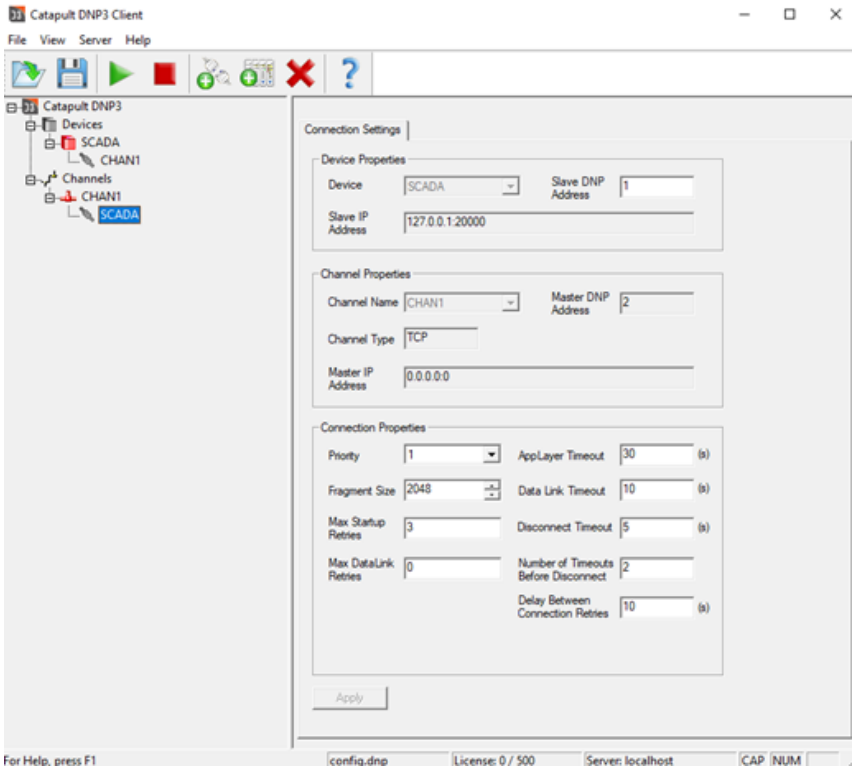


Figure 3.24: DNP driver interface

4 Testing

The final step is testing the configuration before downloading it to the new hardware. During the project individual functionalities were tested such as the controller logic. However, simulation of the RTU and MTU's communication configuration was not possible because the ControlWave Designer does not support virtual communication simulation. Since there was no controller available at the company, testing it has not been done.

Before starting up the system and using it to operate the processing unit two tests must be performed: Factory Acceptance Test or FAT, and Site Acceptance Test or SAT.

FAT as its name implies is a test done before shipping the system to the site to verify that it operates as expected from the hardware and software point of view. In this test all major components are assembled in one manageable area.

The SAT is done at the actual site to verify the reception of all components without being affected by transportation and installation. The test also checks the functionality and performance of the system along with the clarity and completeness of the documentation.

5 Conclusion

This chapter's objectives were to define and implement the control logic and communication protocols, configure SCADA system parameters such as: alarms points and devices; and present the HMI displays and test them.

Conclusion and perspectives

This graduation project, realized within Integration Objects, entitled Oil site telemetry system upgrade was realized in three main phases.

To start, a bibliographic research had to be conducted, in order to understand key aspects about the sites. This phase also includes understanding the client specifications and project planning.

Subsequently, the design phase of the required system was started, in which a functional analysis of the customer specifications was established, and then the hardware needed was identified and sized to achieve the design of an architecture that responds to the objectives set for the project. Engineering documents of the electrical panels and cabinets arrangement were generated during this phase using AutoCAD Electrical as a design software.

Lastly, the work was focused on the automation of the control, and the creation of the supervision systems. Using ControlWave designer to program the controllers and configure the Modbus communication between RTU and MTU. And finally, with the use of Cimplicity, a SCADA system was configured along with the HMI screens and the DNP3 communication configuration.

The system gives access to real time data of the different sites, allowing for early fault detection and risk reduction from a remote location.

During this project, the limitation encountered was the unavailability of equipment which meant the inability to test the program functions and communication configurations.

This project is part of the modernization and automation of facilities based on the industry 3.0 concept, which corresponds to the use of computers and programmable logic

controllers to organize production materials.

A new goal to be achieved in the next project is to use cyber-physical systems (such as smartphones) to control, direct, and monitor the production process. It is the concept of Industry 4.0.

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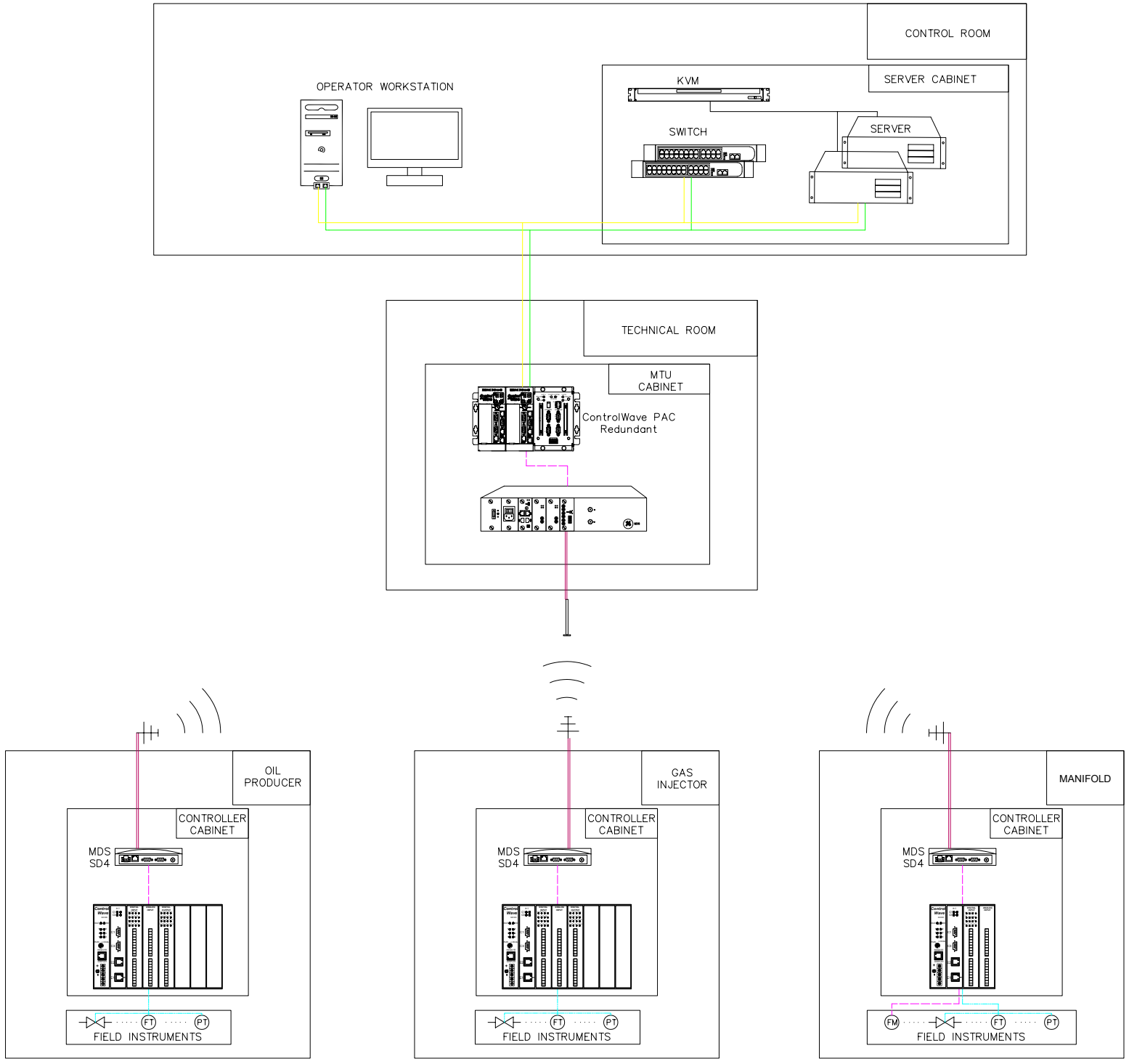
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A

Appendix 1: General Arrangement



Reference Document	Number

NOTES:

LEGEND:

- Serial communication
- Ethernet primary link
- Ethernet secondary link
- Coaxial cable
- Electrical signals
- Radiofrequency
- VGA and USB

Rev	Date	Description	Drawn By	Checked By	Approved By	Client
2	30.03.2021	For Approval	FBF	MAG		
1	23.03.2021	For Approval	FBF	MAG		
0	10.03.2021	For Approval	FBF	MAG		

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

integration objects

Title:
SYSTEM ARCHITECTURE
GENERAL ARCHITECTURE

Integration Objects Document Number :

Format : A4 (297 x 210)

B

Appendix 2: Gas Injector wiring diagram

PROJECT : OIL SITE TELEMETRY SYSTEM UPGRADE

DOCUMENTATION : GAS INJECTOR WELL WIRING DIAGRAMS

CONTRACTOR : INTEGRATION OBJECTS

CLIENT : XXXXX

2	30.04.2021	For Approval	FBF	MAG		
1	21.04.2021	For Approval	FBF	MAG		
0	24.03.2021	For Approval	FBF	MAG		
Rev	Date	Description	Drawn By	Checked By	Approved By	Client

Reference Document	Number

NOTES:

LEGEND:

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1	21.04.2021	For Approval	FBF	MAG		
0	24.03.2021	For Approval	FBF	MAG		
Rev	Date	Description	Draw By	Checked By	Approved By	Client

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

integration objects

Title:
COVER
GAS INJECTOR WELL WIRING DIAGRAMS

Integration Objects Document Number:

Format : A4 (297 x 210)

PAGE	REVISION							DESCRIPTION
	0	1	2					
01	X	X						COVER
02	X	X						INDEX
03	X	X	X					LEGEND & ABBREVIATION
04	X	X	X					GAS INJECTOR WELL RTU CABINET POWER DISTRIBUTION
05	X	X	X					GAS INJECTOR WELL DIGITAL INPUTS TERMINATION
06	X	X						GAS INJECTOR WELL ANALOG INPUTS TERMINATION
07	X	X	X					GAS INJECTOR WELL DIGITAL OUTPUTS TERMINATION

Reference Document	Number

NOTES:

LEGEND:

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1	21.04.2021	For Approval	FBF	MAG		
0	24.03.2021	For Approval	FBF	MAG		

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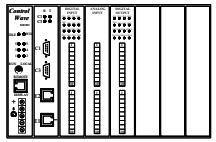
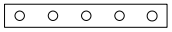
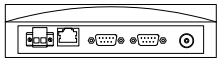
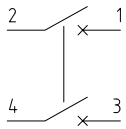
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Title:
INDEX
GAS INJECTOR WELL WIRING DIAGRAMS

Integration Objects Document Number:

Format : A4 (297 x 210)

LEGEND & ABBREVIATION

LEGEND	DESCRIPTION
	Terminal Block
	Fused Terminal Block
	Lightning Protection
	ControlWave Micro 8 slot rack
	Ground Bar
	MDS SD4
	Bipolar Circuit Breaker

ABBREVIATION	DESCRIPTION
TB	TERMINAL BLOCK
DI	DIGITAL INPUT
AI	ANALOG INPUT
DO	DIGITAL OUTPUT
PSSM	POWER SUPPLY SEQUENCER MODULE
RL	RELAY MODULE
MCB	MINIATURE CIRCUIT BREAKER

Reference Document	Number

NOTES:

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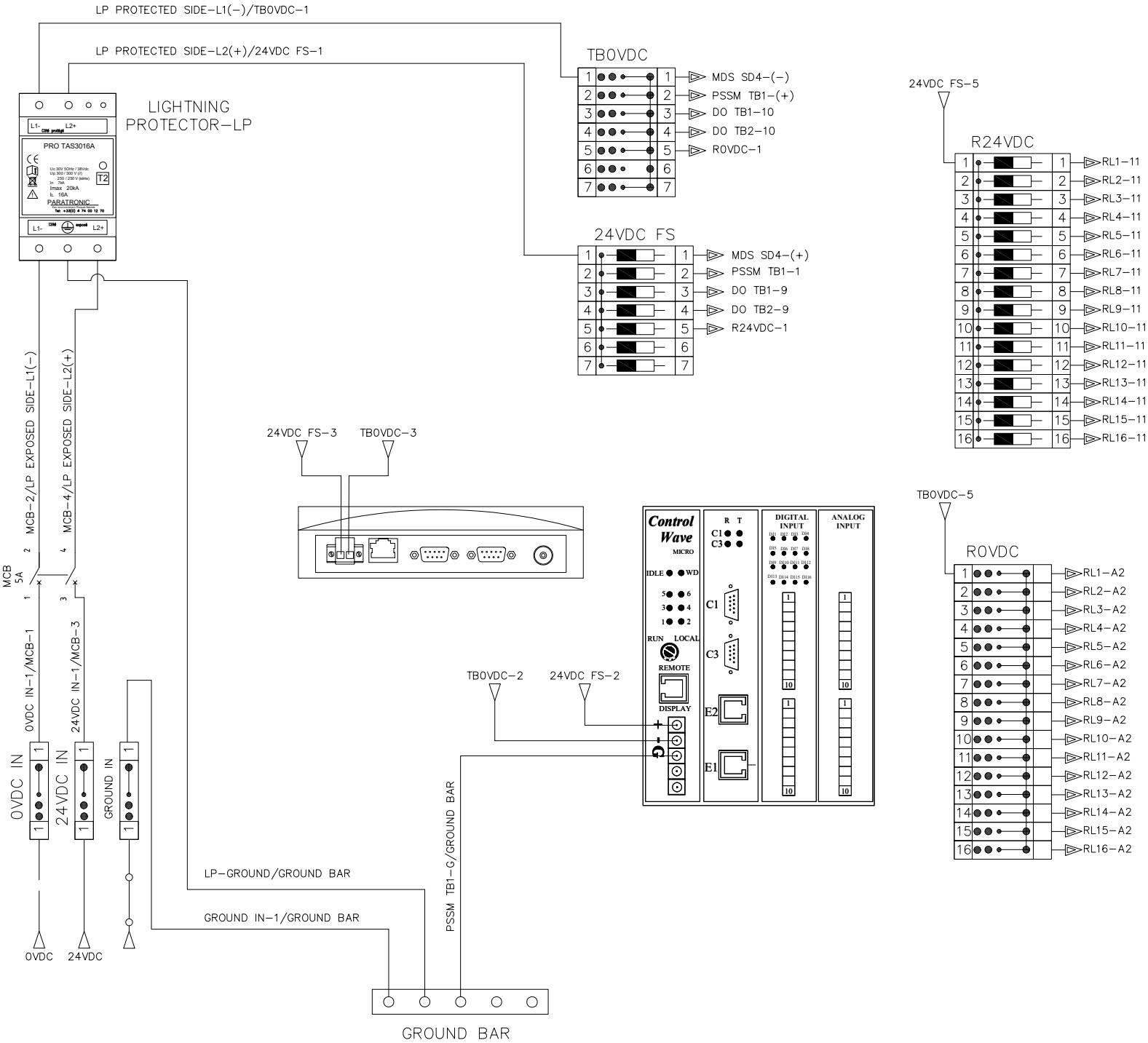
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Title:
LEGEND & ABBREVIATION
GAS INJECTOR WELL WIRING DIAGRAMS

Integration Objects Document Number:

Format : A4 (297 x 210)

GAS INJECTOR WELL RTU CABINET POWER DISTRIBUTION



Reference Document	Number

NOTES:

FUSE	CURRENT RATING (A)
FS1	3
FS2	0.062
FS3	0.6
FS4	0.6
FS5	0.125

All ground wires are 14AWG

Other wires are 16AWG

LEGEND:

Rev	Date	Description	Draw By	Checked By	Approved By	Client
2	30.04.2021	For Approval	FBF	MAG		
1	21.04.2021	For Approval	FBF	MAG		
0	24.03.2021	For Approval	FBF	MAG		

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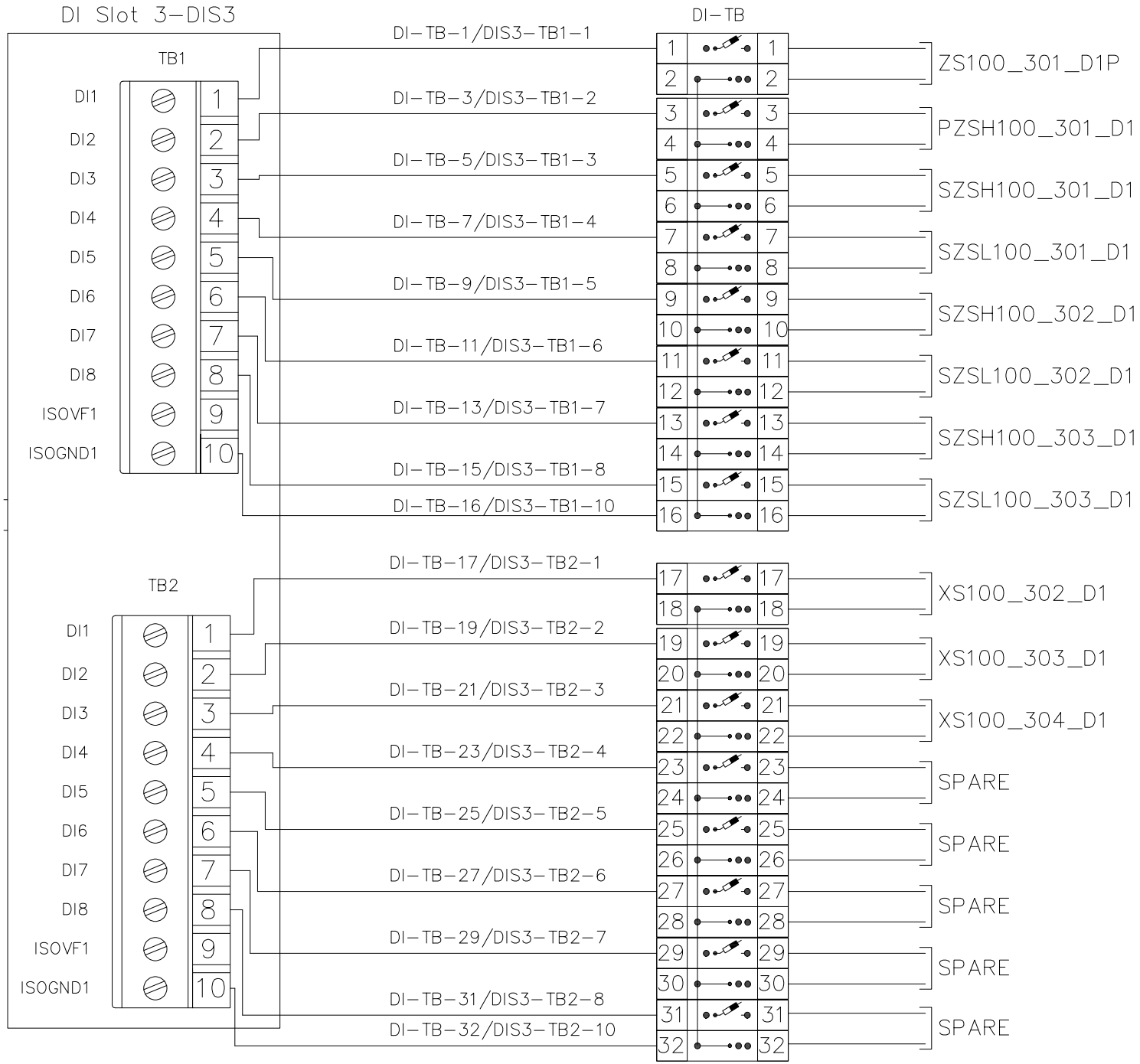
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GAS INJECTOR WELL WIRING DIAGRAMS

Integration Objects Document Number:

Format : A4 (297 x 210)

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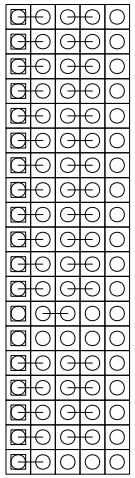


Reference Document	Number

NOTES:

Fuses rated at 0.125A

Jumpers Assignment:



W1 (FOR DI1)
W2 (FOR DI2)
W3 (FOR DI3)
W4 (FOR DI4)
W5 (FOR DI5)
W6 (FOR DI6)
W7 (FOR DI7)
W8 (FOR DI8)
W9 (FOR DI9)
W10 (FOR DI10)
W11 (FOR DI11)
W12 (FOR DI12)
W13 (FOR LEDS)
W14 (FACTORY USE)
W15 (FOR DI13)
W16 (FOR DI14)
W17 (FOR DI15)
W18 (FOR DI16)
W19 (FOR VIN)

LEGEND:

Rev	Date	Description	Draw By	Checked By	Approved By	Client
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1	21.04.2021	For Approval	FBF	MAG		
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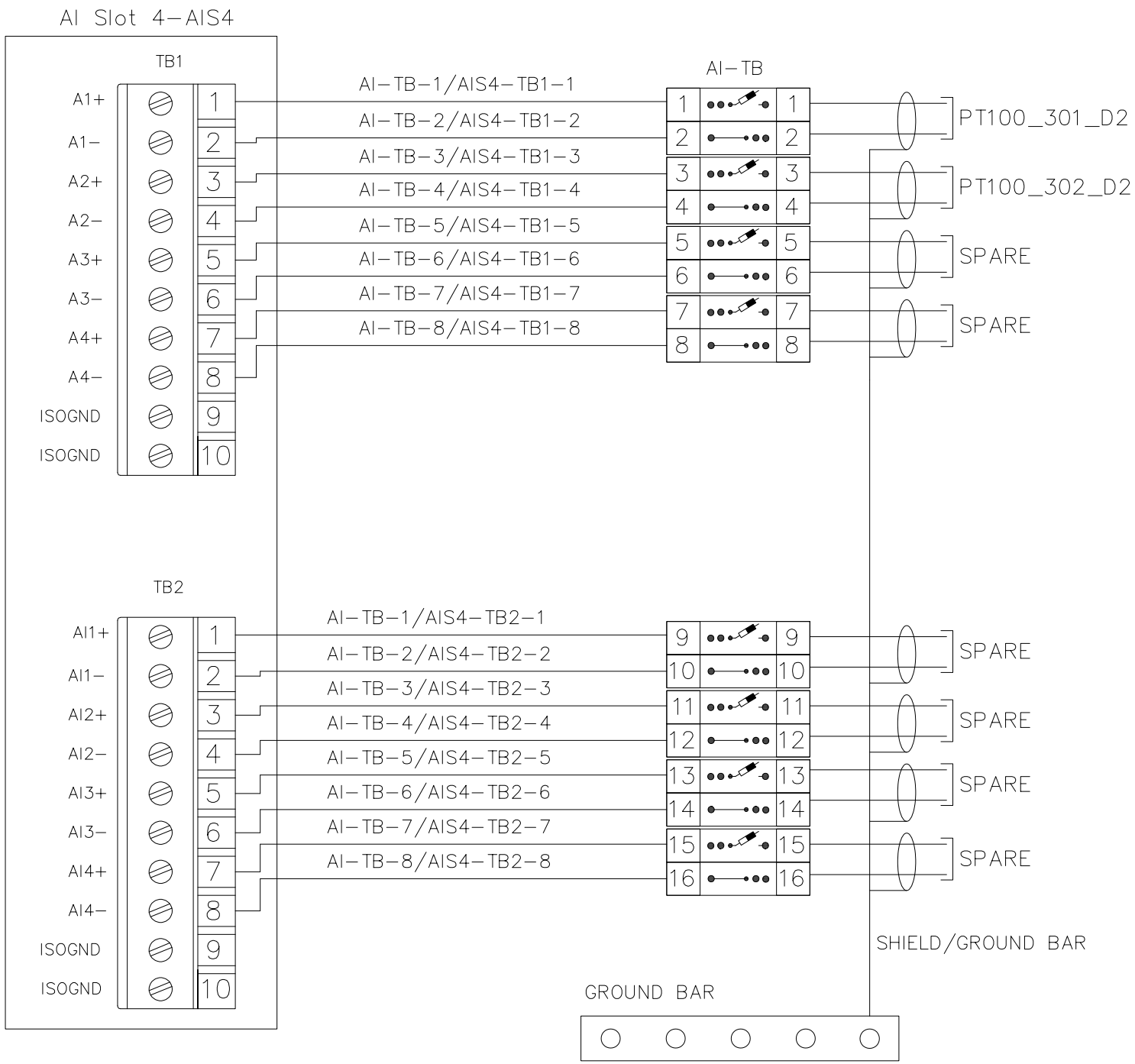
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Integration Objects Document Number:

Format : A4 (297 x 210)

GAS INJECTOR WELL ANALOG INPUTS TERMINATION

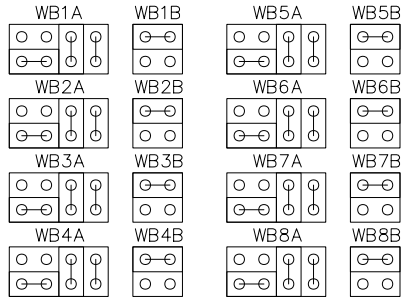


Reference Document	Number

NOTES:

Fuses rated at 0.125A

Jumpers Assignment:



LEGEND:

Rev	Date	Description	Draw By	Checked By	Approved By	Client
2	30.04.2021	For Approval	FBF	MAG		
1	21.04.2021	For Approval	FBF	MAG		
0	24.03.2021	For Approval	FBF	MAG		

Project: OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

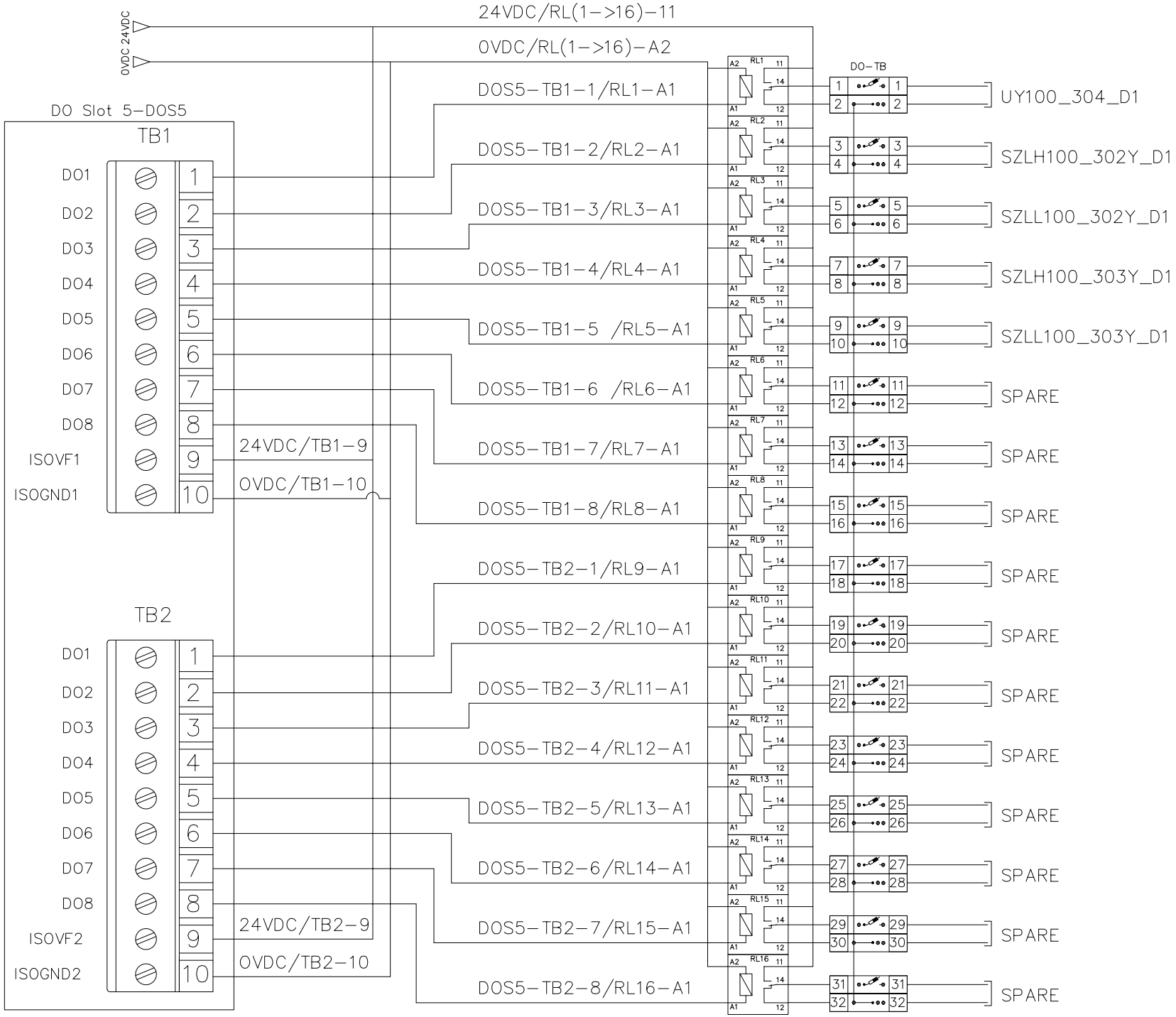
integration objects

Title: GAS INJECTOR WELL ANALOG INPUTS TERMINATION
GAS INJECTOR WELL WIRING DIAGRAMS

Integration Objects Document Number:

Format : A4 (297 x 210)

GAS INJECTOR WELL DIGITAL OUTPUTS TERMINATION



Reference Document	Number

NOTES:

Fuses rated at 0.125A

LEGEND:

Rev	Date	Description	Draw By	Checked By	Approved By	Client
2	30.04.2021	For Approval	FBF	MAG		
1	21.04.2021	For Approval	FBF	MAG		
0	24.03.2021	For Approval	FBF	MAG		

Project: OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

integration objects

Title: GAS INJECTOR WELL DIGITAL OUTPUTS TERMINATION
GAS INJECTOR WELL WIRING DIAGRAMS

Integration Objects Document Number:

Format : A4 (297 x 210)

C

Appendix 3: Gas Injector cabinet

PROJECT : OIL SITE TELEMETRY SYSTEM UPGRADE

DOCUMENTATION : GAS INJECTOR WELL RTU CABINET LAYOUT

CONTRACTOR : INTEGRATION OBJECTS

CLIENT : XXXXX

2	03.05.2021	For Approval	FBF	MAG		
1	23.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		
Rev	Date	Description	Drawn By	Checked By	Approved By	Client

Reference Document	Number

NOTES:

LEGEND:

2	03.05.2021	For Approval	FBF	MAG		
1	23.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		
Rev	Date	Description	Drawn By	Checked By	Approved By	Client

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

integration objects

Title:
COVER
GAS INJECTOR WELL RTU CABINET LAYOUT

Integration Objects Document Number :

Format : A4 (297 x 210)

PAGE	REVISION							DESCRIPTION
	0	1	2					
01	X	X						COVER
02	X	X						INDEX
03	X	X						LEGEND & ABBREVIATION
04	X	X						GAS INJECTOR WELL RTU CABINET VIEWS
05	X	X	X					GAS INJECTOR WELL RTU CABINET GENERAL ARRANGEMENT
06	X	X	X					NOMENCLATURE

Reference Document	Number

NOTES:

LEGEND:

Rev	Date	Description	Drawn By	Checked By	Approved By	Client
2	03.05.2021	For Approval	FBF	MAG		
1	23.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		

Project:
UPGRADING OF A TELEMETRY SYSTEM AT AN OIL SITE

Client:

integration objects

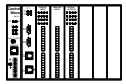
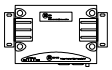
Title:
INDEX
GAS INJECTOR WELL RTU CABINET LAYOUT

Integration Objects Document Number :

Format : A4 (297 x 210)

LEGEND & ABBREVIATION

LEGEND:

Symbol	Description
	CONTROLWAVE Micro 8 slots rack
	MDS SD4
	Ground Bar
	Feed–Through
	Fused terminal block
	Lightning protector
	Bipolar circuit breaker
	Relay

Reference Document	Number

NOTES:

LEGEND:

Rev	Date	Description	Drawn By	Checked By	Approved By	Client
2	03.05.2021	For Approval	FBF	MAG		
1	23.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

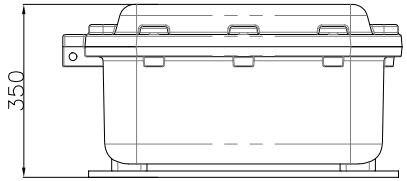
integration objects

Title:
LEGEND AND ABBREVIATION
GAS INJECTOR WELL RTU CABINET LAYOUT

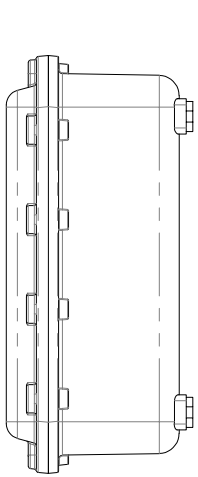
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Format : A4 (297 x 210)

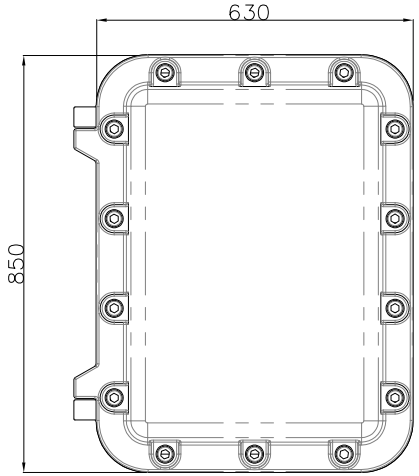
GAS INJECTOR WELL RTU CABINET VIEWS



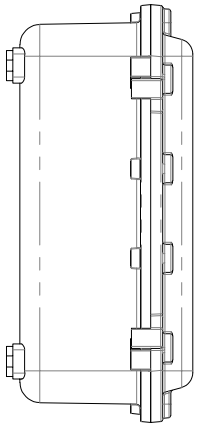
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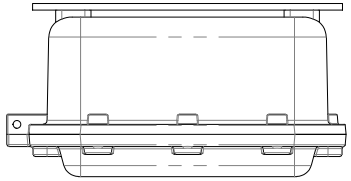
RIGHT



FRONT



LEFT



TOP

Reference Document	Number

NOTES:

LEGEND:

Rev	Date	Description	Drawn By	Checked By	Approved By	Client
2	03.05.2021	For Approval	FBF	MAG		
1	23.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

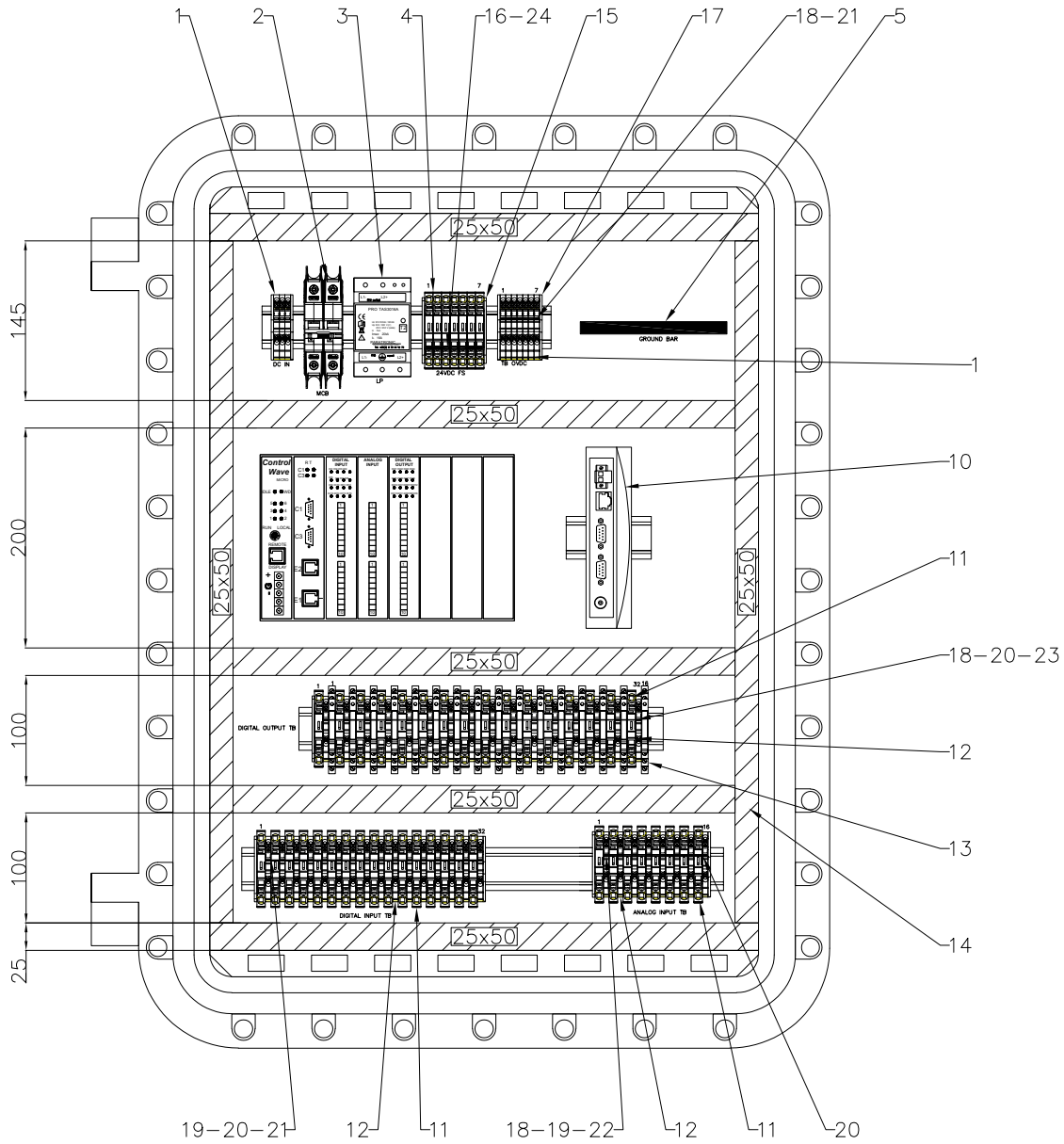
integration objects

Title:
GAS INJECTOR WELL RTU CABINET VIEWS
GAS INJECTOR WELL RTU CABINET LAYOUT

Integration Objects Document Number :

Format : A4 (297 x 210)

GAS INJECTOR WELL RTU CABINET GENERAL ARRANGEMENT



Reference Document	Number

NOTES:

LEGEND:

Rev	Date	Description	Drawn By	Checked By	Approved By	Client
2	03.05.2021	For Approval	FBF	MAG		
1	23.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

integration objects

Title:
GAS INJECTOR WELL RTU CABINET GENERAL ARRANGEMENT
GAS INJECTOR WELL RTU CABINET LAYOUT

Integration Objects Document Number :

Format : A4 (297 x 210)

NOMENCLATURE

ITEM	QUANTITY	PART NUMBER	DESCRIPTION	MANUFACTURER	EXISTING/NEW
1	10	0115271.22	Feed Through terminal block	ENTRELEC	EXISTING
2	1	1489-M	Circuit breaker 2P	ROCKWELL	NEW
3	1	PRO TAS3016A	Lightning Protector	PARATRONIC	EXISTING
4	7	0115657.25	Fused terminal block	ENTRELEC	EXISTING
5	1		GROUND BAR		EXISTING
6	1	396563-06-6	150 MHz CPU, 2M SR, 16M FLASH, 2-ENET & 1-RS 232 &1-RS 485 (CE Approved)	EMERSON	NEW
7	1	396571-02-6	16 Digital Input module	EMERSON	NEW
8	1	396604-01-3	8 Analog Inputs module	EMERSON	NEW
9	1	396572-02-2	16 Digital Outputs module	EMERSON	NEW
10	1	SD04MD-AES-NNSNN	MDS SD4 ethernet and serial, 350-400MHz	GE	NEW
11	24	3211874	IO Fused terminal block	PHOENIX CONTACT	NEW
12	24	3213968	IO Feed Through terminal block	PHOENIX CONTACT	NEW
13	16	2966171	Output Relays	PHOENIX CONTACT	NEW
14	3.2m	30450024	Cable Trunking 25x75mm	BETADUCT	NEW
15	5	3022276	Fused terninal block end clamp	PHOENIX CONTACT	NEW
16	1	3030310	Fused terminal block plug-in bridge 5 positions	PHOENIX CONTACT	NEW
17	6	1016672	Feed through terminal block end cover	PHOENIX CONTACT	NEW
18	2	3030190	Feed through terminal block plug-in bridge 5 positions	PHOENIX CONTACT	NEW
19	2	3030213	Feed through terminal block plug-in bridge 10 positions	PHOENIX CONTACT	NEW
20	3	3030226	Feed through terminal block plug-in bridge 20 positions	PHOENIX CONTACT	NEW
21	2	3030174	Feed through terminal block plug-in bridge 3 positions	PHOENIX CONTACT	NEW
22	1	3030161	Feed through terminal block plug-in bridge 2 positions	PHOENIX CONTACT	NEW
23	1	3030187	Feed through terminal block plug-in bridge 4 positions	PHOENIX CONTACT	NEW
24	1	3030297	Fused terminal block plug-in bridge 3 positions	PHOENIX CONTACT	NEW

Reference Document	Number

NOTES:

LEGEND:

Rev	Date	Description	Drawn By	Checked By	Approved By	Client
2	03.05.2021	For Approval	FBF	MAG		
1	23.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

integration objects

Title:
NOMENCLATURE
GAS INJECTOR WELL RTU CABINET LAYOUT

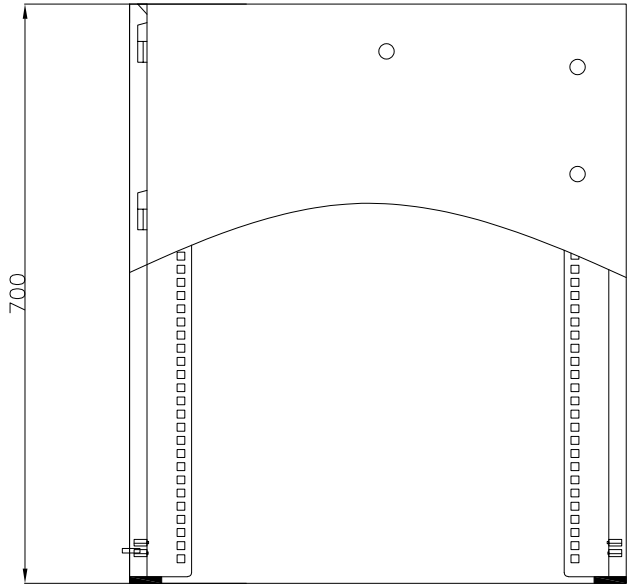
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Format : A4 (297 x 210)

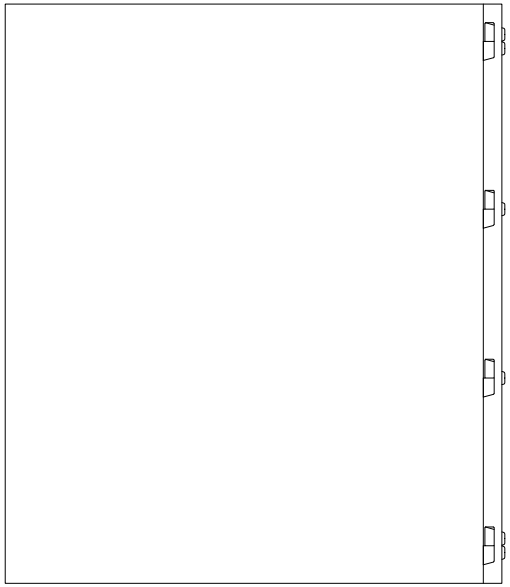
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Appendix 4: MTU cabinet

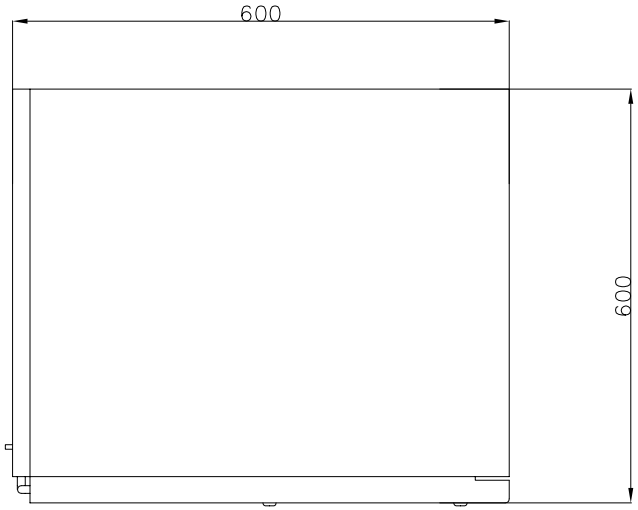
MTU CABINET VIEWS



FRONT



LEFT



TOP

Reference Document	Number

NOTES:

LEGEND:

Rev	Date	Description	Drawn By	Checked By	Approved By	Client
3	07.05.2021	For Approval	FBF	MAG		
2	03.05.2021	For Approval	FBF	MAG		
1	23.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

integration objects

Title:
MTU CABINET VIEWS
MTU CABINET LAYOUT

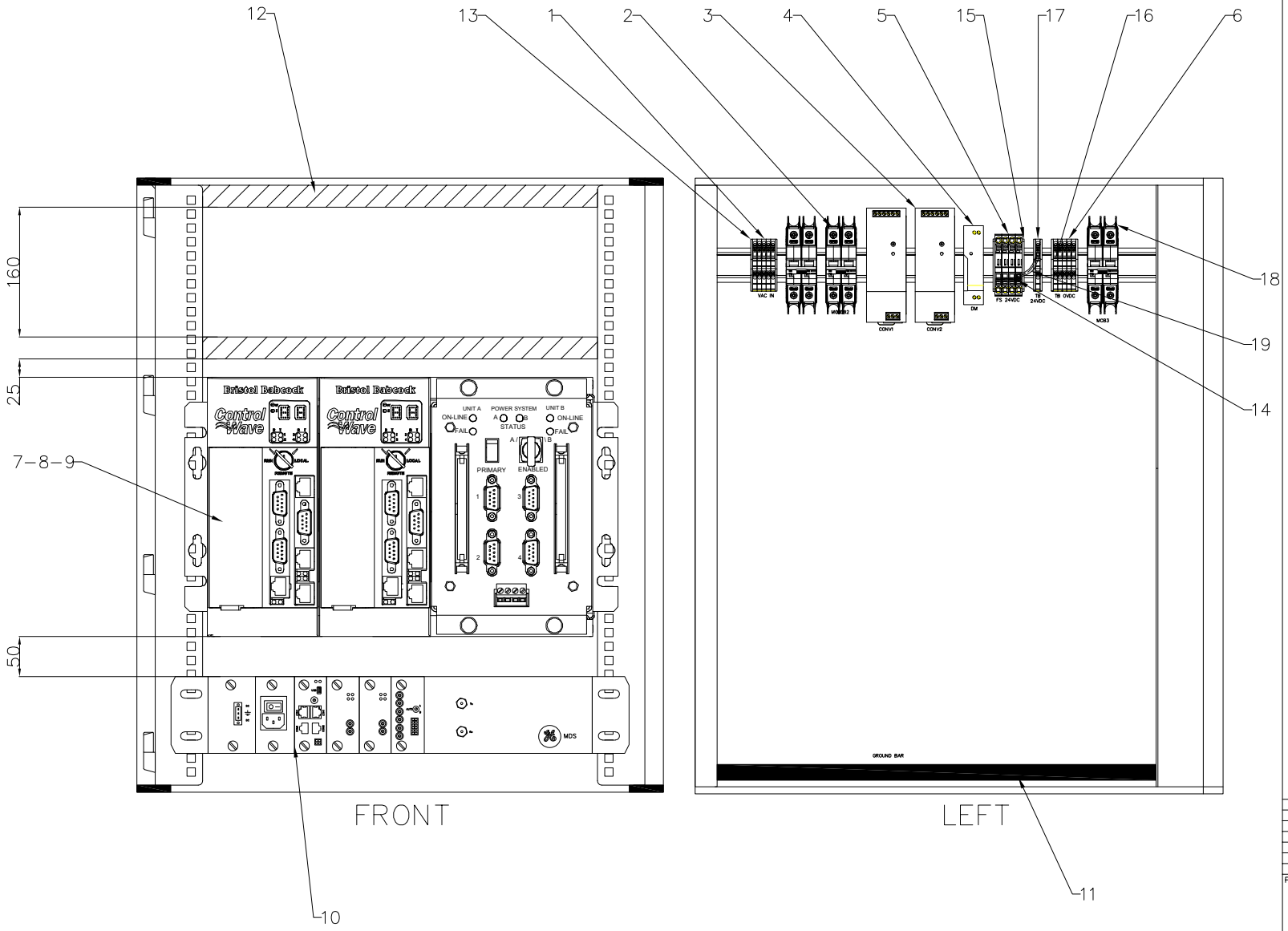
Integration Objects Document Number :

Format : A4 (297 x 210)

MTU CABINET GENERAL ARRANGEMENT

Reference Document	Number

NOTES:



LEGEND:

Rev	Date	Description	Drawn By	Checked By	Approved By	Client
3	07.05.2021	For Approval	FBF	MAG		
2	03.05.2021	For Approval	FBF	MAG		
1	23.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

integration objects

Title:
MTU CABINET GENERAL ARRANGEMENT
MTU CABINET LAYOUT

Integration Objects Document Number :

Format : A4 (297 x 210)

NOMENCLATURE

ITEM	QUANTITY	PART NUMBER	DESCRIPTION	MANUFACTURER	EXISTING/NEW
1	5	3213968	Feed Through terminal block	PHOENIX CONTACT	NEW
2	2	A9F90282	Circuit breaker 2P 12.5A	SCHNEIDER ELECTRIC	NEW
3	2	1096432	230VAC/24VDC Converter 10A	PHOENIX CONTACT	NEW
4	2	2905489	10A Diode module	PHOENIX CONTACT	NEW
5	4	3211874	24VDC Fused terminal block	PHOENIX CONTACT	NEW
6	5	3213968	0VDC Feed Through terminal block	PHOENIX CONTACT	NEW
7	2	CWREDCPU – A–4	PSSM 24VDC	EMERSON	NEW
8	2	CWREDCPU – BC–12	CPU 2 RS–232 3 Ethernet	EMERSON	NEW
9	1	CWREDCPU –D–1	CCRSM 2 RS–232 2 RS–485	EMERSON	NEW
10	1	MPRS4ARDNS1S2N11NNSNN	MDS Master Station	GE	NEW
11	1		Ground Bar		NEW
12		30450024	Cable Trunking 25x75mm	BETADUCT	NEW
13	2	1016672	Feed through terminal block end cover	PHOENIX CONTACT	NEW
14	1	3030307	Fused terminal block plug–in bridge 4 positions	PHOENIX CONTACT	NEW
15	2	3022276	Fused terninal block end clamp	PHOENIX CONTACT	NEW
16	1	3030187	Feed through terminal block plug–in bridge 4 positions	PHOENIX CONTACT	NEW
17	1	3213968	24VDC Feed Through terminal block	PHOENIX CONTACT	NEW
18	1	1489–M2D060	Circuit breaker 2P 6A	ROCKWELL	NEW
19	1	3030170	Wire bridge 60mm	PHONENIX CONTACT	NEW

Reference Document	Number

NOTES:

LEGEND:

Rev	Date	Description	Drawn By	Checked By	Approved By	Client
3	07.05.2021	For Approval	FBF	MAG		
2	03.05.2021	For Approval	FBF	MAG		
1	23.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

integration objects

Title:
NOMENCLATURE
MTU CABINET LAYOUT

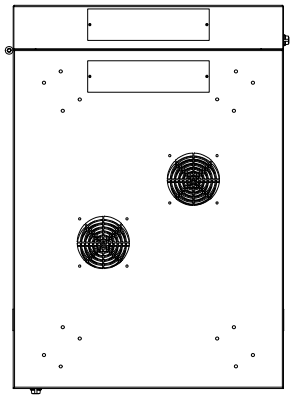
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Format : A4 (297 x 210)

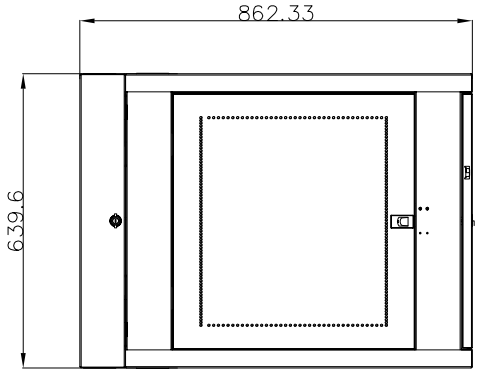
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Appendix 5: Server cabinet

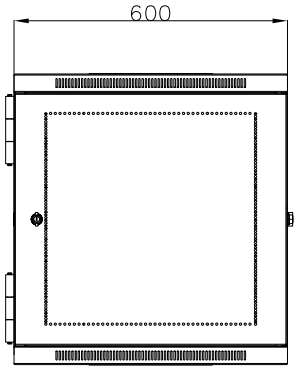
SERVER CABINET VIEWS



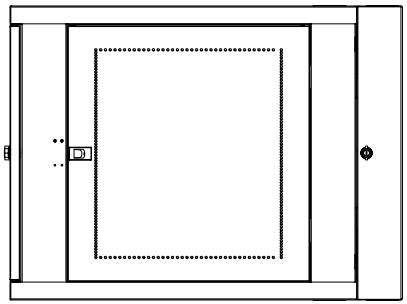
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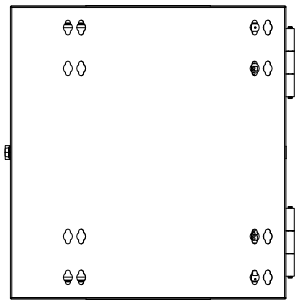
RIGHT



FRONT



LEFT



BACK

Reference Document	Number

NOTES:

LEGEND:

Rev	Date	Description	Drawn By	Checked By	Approved By	Client
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2	03.05.2021	For Approval	FBF	MAG		
1	26.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

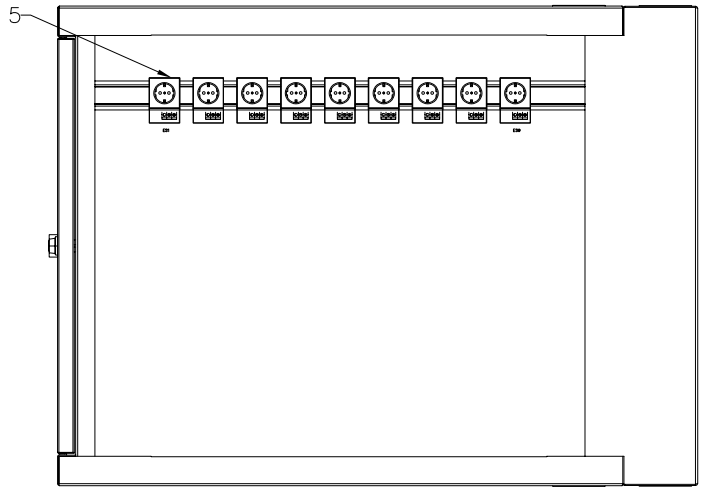
integration objects

Title:
SERVER CABINET VIEWS
SERVER CABINET LAYOUT

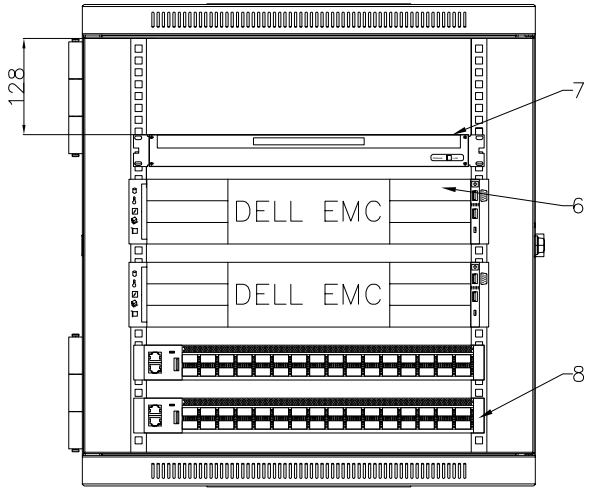
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Format : A4 (297 x 210)

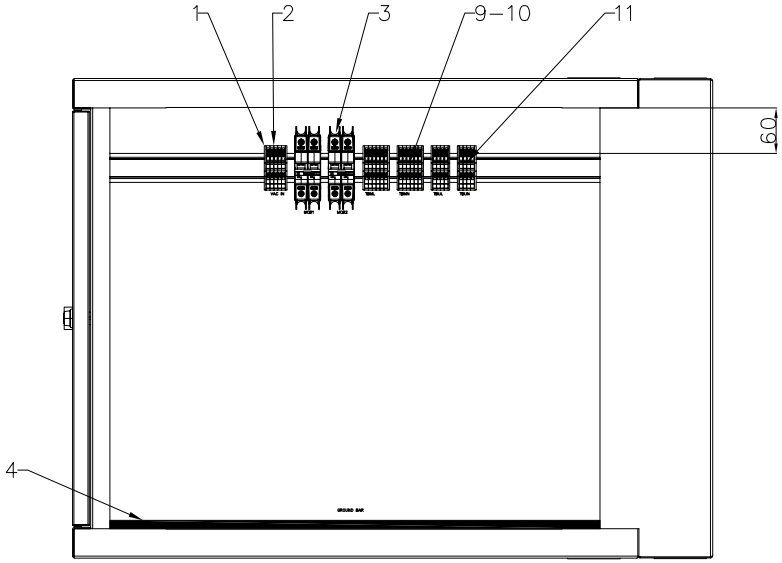
SERVER CABINET GENERAL ARRANGEMENT



RIGHT



FRONT



LEFT

Reference Document	Number

NOTES:

LEGEND:

Rev	Date	Description	Drawn By	Checked By	Approved By	Client
3	07.05.2021	For Approval	FBF	MAG		
2	03.05.2021	For Approval	FBF	MAG		
1	26.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

integration objects

Title:
SERVER CABINET GENERAL ARRANGEMENT
MTU CABINET LAYOUT

Integration Objects Document Number :

Format : A4 (297 x 210)

NOMENCLATURE

ITEM	QUANTITY	PART NUMBER	DESCRIPTION	MANUFACTURER	EXISTING/NEW
1	10	1016672	Feed through terminal block end cover	PHOENIX CONTACT	NEW
2	4	3213968	Feed Through terminal block	PHOENIX CONTACT	NEW
3	2	1489-M2C200	Circuit breaker 2P 20A	ROCKWELL AUTOMATION	NEW
4	1		Ground Bar		NEW
5	9	03501.0-01	Electrical socket	STEGO	NEW
6	2		PowerEdge R540 Rack Server	DELL	NEW
7	1	B040-016-19	Rack-mount KVM	TRIPP LITE	NEW
8	2	C9500-32C	Switch	CISCO	NEW
9	2	3030190	Feed through terminal block plug-in bridge 5 positions	PHOENIX CONTACT	NEW
10	2	3030161	Feed through terminal block plug-in bridge 2 positions	PHOENIX CONTACT	NEW
11	2	3030187	Feed through terminal block plug-in bridge 4 positions	PHOENIX CONTACT	NEW

Reference Document	Number

NOTES:

LEGEND:

Rev	Date	Description	Drawn By	Checked By	Approved By	Client
3	07.05.2021	For Approval	FBF	MAG		
2	03.05.2021	For Approval	FBF	MAG		
1	23.04.2021	For Approval	FBF	MAG		
0	12.04.2021	For Approval	FBF	MAG		

Project:
OIL SITE TELEMETRY SYSTEM UPGRADE

Client:

integration objects

Title:
NOMENCLATURE
SERVER CABINET LAYOUT

Integration Objects Document Number :

Format : A4 (297 x 210)

العنوان: ترقية نظام القياس عن بعد لموقع النفط

ملخّص: الهدف الرئيسي من هذا المشروع هو ترقية نظام القياس عن بعد في موقع نفط. في هذا التقرير نصف الخطوات اللازمة لتنفيذ النظام. بدءًا من اختيار المواد وتحجيمها، ثم توليد مخططات الأسلاك الكهربائية. والانتهاء بتنفيذ منطق التحكم، وتطوير، واجهات الانسان والألة.

المفاتيح: نظام القياس عن بعد، الوحدة الطرفية البعيدة، وحدة القياس عن بعد الرئيسية، الوحدة الطرفية البعيدة واستحصال البيانات، مودبس، دي إن بي 3، كونترول وايف ميكرو، كونترول وايف راد

Titre : Mise à niveau d'un système de télémétrie d'un site pétrolier

Résumé : Le but principal de ce projet est la mise à niveau du système de télémétrie d'un site pétrolier. Dans ce rapport nous exposons les étapes nécessaires pour l'implémentation du système. En commençant par la sélection et le dimensionnement du matériel, suivi par la génération des diagrammes de câblage électrique. Pour finir avec l'implémentation de la logique de commande et le développement des interfaces Homme-Machine.

Mots-clés : Système de télémétrie, RTU, MTU, SCADA, Modbus, DNP3, ControlWave Micro, ControlWave RED

Title : Oil site telemetry system upgrade

Abstract : The main goal of this project is the upgrade of the telemetry system at an oil site. In this report we describe the necessary steps for implementing the system. Starting with the hardware selection and sizing, followed by the generation of the wiring diagrams and schematics. To finally end with the control logic configuration and the Human-Machine interfaces development.

Key Words: Telemetry system, RTU, MTU, SCADA, Modbus, DNP3, ControlWave Micro, ControlWave RED

Company information

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Email : marketing@integrationobjects.com